

Quasi-Experimental Analysis Reveals Neuro-Genetic Susceptibility to Neighborhood Socioeconomic Adversity in Children's Psychotic-Like Experiences

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Abstract

Socioeconomic deprivation is linked to psychiatric vulnerability in children, yet the sources of individual variability remain unclear. Using an instrumental-variable random-forest framework (IV Forest) in a longitudinal cohort of 2,135 children, we estimate the association between neighborhood socioeconomic deprivation and delay discounting and psychotic-like experiences (PLEs) using variation in neighborhood adversity induced by state source-of-income anti-discrimination laws. Under standard IV assumptions, higher neighborhood adversity relates to steeper delay discounting and higher PLEs on average, and these associations vary systematically across children. These findings support the behavioral poverty trap framework, which is a self-reinforcing cycle where poverty-induced psychological distress relates to decisions that may perpetuate economic hardship. Moving beyond average relationships, an integrated IV Forest model that combines delay discounting, prefrontal-limbic morphometry, reward-task activation, and polygenic scores delineates marked significant heterogeneity in associations between neighborhood adversity and PLEs across children. Children most vulnerable to neighborhood adversity showed a paradoxical pattern: higher genetic predisposition for cognitive achievement (GPS for cognitive performance, IQ, and educational attainment) combined with specific limbic alterations (reduced volumes, heightened reward-related activation). These findings challenge diathesis-stress models and support differential susceptibility theory. By identifying who is most vulnerable to neighborhood disadvantage and why, our results inform precision-medicine approaches to preventing childhood psychopathology and breaking cycles of socioeconomic disadvantage.

Introduction

A child's environment is a powerful predictor of their lifelong health and economic outcomes. Adverse childhood environments, such as low family income, malnutrition, physical or sexual abuse, and unsafe neighborhoods, are linked to an heightened risk of various mental or physical health issues, including psychosis (1-3), impoverished cognitive ability (1-3), anxiety, bipolar disorder, self-harm, depression (4-6), substance abuse, and obesity (2, 3, 7). Furthermore, these environments are associated with negative social outcomes, such as poor academic performance (8, 9), low income, unemployment (10, 11), higher rates of imprisonment, and increased likelihood of teen pregnancy (12, 13). Additionally, exposure to these adverse conditions in childhood is associated with a propensity for engaging in risky behaviors, including criminal activity (13, 14), excessive consumption of calorie-dense foods (15), substance use (16, 17), deficient self-control (18), and disrupted reward processing (19).

One prominent framework, the “behavioral poverty trap,” posits that exposure to socioeconomic deprivation alters reward valuation, promoting impulsive decision-making that can lead to a range of suboptimal behavior and outcomes (20, 21). Individuals with steeper discounting of future rewards (i.e., value present rewards much higher than future rewards) are more likely to save less, invest less in their education, engage in criminal activities, exhibit lower academic performance, and accumulate less economic wealth (20, 22-24). Such impairments in intertemporal valuation are not only associated with financial and social disadvantages but are also linked to psychiatric disorders, including psychosis, attention deficit/hyperactivity disorder (ADHD), and addiction (25, 26).

In this paper, we focused on psychotic-like experiences (PLEs) as a key psychiatric outcome. Around 17% of 9-12 years old children report PLEs (27), individuals with PLEs at

age 11 had greater risk of developing psychosis in adulthood (28, 29). Prior studies revealed that PLEs are correlated to heightened vulnerability to other psychopathologies (30) including suicidal behavior (1), mood, anxiety, and substance disorders (31-33) and exhibit the strongest association with environmental risk factors in comparison to other internalizing/externalizing symptoms during adolescence (2).

The link between deprivation and PLEs is likely moderated by core neurodevelopmental processes, and we propose that delay discounting provides a critical window into these mechanisms (31, 34, 35). Delay discounting, which is evidenced by the extent to which individuals' discount future rewards, pertains to their intertemporal decision-making and impulsive behavior. This measure of intertemporal choice is not merely a behavioral quirk but a quantifiable index of reward valuation governed by prefrontal-limbic circuits (25, 26). Children who experienced social adversities show steeper discounting of future rewards in adulthood (36, 37) and have higher PLEs (1, 2, 38). Comparative studies have demonstrated that individuals with psychosis discount future rewards more steeply than healthy controls (31, 34, 35), a pattern that is uniquely pronounced in psychosis compared to other psychopathologies such as primary mood disorders (34), major depressive disorder (34), and bipolar disorder (35). Crucially, the same mesocorticolimbic dopaminergic systems that regulate reward anticipation are heavily implicated in the pathophysiology of psychosis (1, 2, 38). This shared neurobiological foundation suggests that individual differences in these reward circuits could be a key mechanism determining how environmental stress translates into psychiatric symptoms.

This raises the central puzzle in developmental science: why do some children exhibit remarkable resilience to adversity while others are exquisitely vulnerable? Answering this requires dissecting individual-level heterogeneity, which likely arises from a complex interplay

of genetic predispositions and neurodevelopmental factors (39-41). Traditional linear models are ill-equipped for this task, as they struggle to capture the high-dimensional, nonlinear gene-environment interactions that likely define a child's vulnerability (42, 43).

Here, we address this challenge by applying a causal machine learning framework to rich, longitudinal, multi-modal data—including genomics, structural and functional MRI, and behavior—from children aged 9-12 years old in the Adolescent Brain Cognitive Development (ABCD) Study. Using instrumental variable (IV) random forests (henceforth IV Forest) (30, 44), we move beyond population-level associations to characterize the specific neuro-genetic patterns associated with vulnerability to the psychiatric risks of socioeconomic adversity. We focus specifically on neighborhood socioeconomic deprivation because it represents a critical environmental layer beyond individual family status. Neighborhood adversities during childhood have been associated with maladaptive neurocognitive development (6, 45, 46), psychiatric disorders (28, 29), and unfavorable social outcomes such as decreased income, reduced probability of college attendance, and limited employment opportunities (47, 48). This phenomenon is particularly pronounced in societies where housing discrimination based on family income or race/ethnicity restricts underprivileged families from selecting neighborhoods that present greater opportunities for upward social mobility, as observed in the United States (47).

This study first estimates the IV-based association between neighborhood socioeconomic deprivation, steeper delay discounting, and elevated PLEs. We then document heterogeneity in this link and observe a multi-modal pattern of vulnerability characterized by specific genetic predispositions, and limbic system structures and functions. Within this pattern, we found a paradoxical finding: a higher genetic predisposition for cognitive achievement is positively—rather than negatively—related to the psychiatric risks of a deprived socioeconomic

130 environment.

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Methods

Study Participants

The ABCD Study recruited participants from 21 research sites across the nation, utilizing a stratified, probability sampling method to capture the sociodemographic variation of the US population (49). We used the baseline, first year, and second year follow-up datasets included in ABCD Release 4.0, downloaded on February 10, 2022.

To ensure the highest quality of data and internal validity for our analyses, we implemented the standardized preprocessing and quality control pipelines recommended by the ABCD Study consortium (50, 51). Of the initial 11,876 ABCD samples, participants not meeting the ABCD Study's MRI quality control standards (N=3,275) were excluded. Similarly, for the delay discounting task, we applied the Johnson-Bickel validation criterion (52), as recommended by the ABCD behavioral workgroup (51), to exclude participants with inconsistent responses (i.e., indifferent point for a given delay larger than that of an indifference point for a longer delay) (N=5,293). We also removed participants without genotype data, MRI data, NIH Toolbox Cognitive Battery, delay discounting, residential address, ADI, and PLEs (N=1,173). Missing values of covariates were imputed using k-nearest neighbors. The final samples included 2,135 children from a variety of race/ethnic groups (**Fig. 1**).

Data

Our primary exposure was neighborhood socioeconomic adversity, measured at baseline using the national percentile scores of the Area Deprivation Index (ADI) (2, 45). The primary outcomes were delay discounting, a model-free measure of reward impulsivity assessed at the

1-year follow-up (51), and PLEs, measured using the Prodromal Questionnaire-Brief Child version at 1- and 2-year follow-ups (53). A lower value of delay discounting denotes steeper discounting of future rewards; a higher value of PLE score indicates greater severity of PLEs. To identify moderators of vulnerability, we utilized a rich, multi-modal dataset including genome-wide polygenic scores (GPS) for cognitive and psychiatric traits, structural MRI measures of brain morphometry (50), and functional MRI data from a monetary incentive delay (MID) task (50). Detailed descriptions of all measures, tasks, and MRI processing pipelines are provided in the **Supplementary Information**.

To adjust for the potential confounding effects, sociodemographic covariates were included. Consistent with existing research on psychiatric disorders in ABCD samples (2, 53-55), we controlled for the child's sex, age, race/ethnicity, caregiver's relationship to a child, BMI, parental education, marital status of the caregiver, household income, parent's age, and family history of psychiatric disorders. The family history of psychiatric disorders, measured as the proportion of first-degree relatives who experienced psychosis, depression, mania, suicidality, previous hospitalization, or professional help for mental health issues (2) was included as a covariate. Given that delay discounting and PLEs are associated with an individual's neurocognitive capabilities (56-58), NIH Toolbox total intelligence was used as a covariate. All covariates were from baseline year observations.

Statistical Analyses

Given the non-randomized, observational nature of the ABCD Study, potential confounding factors, such as genetic, environmental variables, and their unobserved common causes, can lead to biased estimations (59). A powerful and effective way to adjust such bias is

the IV regression. Designed to isolate a quasi-random source of variation in the treatment/exposure variable, IV regression is an effective method to conduct quasi-experimental analysis using non-randomized, observational data for research in various domains, including economics (60, 61), psychology (62), neuroscience (63), and psychiatry (64).

Our instrument is the presence of state-level source of income (SOI) laws at baseline year assessment, which prohibit housing discrimination against voucher holders. These laws increase the landlords' acceptance of housing vouchers, aiding low-income families in securing quality housing. As such, SOI laws are critical in enabling better residential environments by mitigating neighborhood socioeconomic adversity. That is, the implementation of these laws provides a *natural experiment*: it strongly predicts a family's neighborhood adversity (i.e., ADI) but is plausibly unrelated to the unobserved confounding factors that typically bias estimates. The IV analysis leverages this exogenous variation from SOI laws to estimate the associations of ADI with delay discounting and PLEs while mitigating bias from unobserved confounders (see **Fig. 2A**). We present theoretical explanations of the IV method and detailed justification of SOI laws as valid instrument in the **Supplementary Information**.

We used IV Forest (65, 66)—a random forest-based IV regression (60)—to adjust for unobserved confounding bias in identifying the potential relationship of ADI with delay discounting and PLEs. The IV Forest method provides nonparametric, doubly robust estimates of the average (group-level) and heterogeneous (individual-level) associations of ADI with these outcomes, interpreted under the IV assumptions (see **Fig. 2B**). This method is noted for delivering estimates with significantly lower mean-squared error compared to conventional k-nearest neighbor methods (65, 66). Furthermore, its use of independent subsamples for model construction and validation ensures honest, overfitting-resistant estimates of average and

heterogeneous relationships (65-67). Notably, this method is particularly useful for analyzing the complex, nonlinear interactions between genetic and environmental factors and their link with neurocognitive development and PLEs, even within the confines of observational data (65). We used cluster-robust inference with the ABCD Study site as the clustering unit for IV Forest estimates. The technical description of the IV Forest algorithm is detailed in the **Supplementary Information**.

Fig. 1 presents the analytical framework of our study, examining the associations of neighborhood socioeconomic adversity with children's decision-making and mental health. ADI, recorded in the baseline year, serves as an indicator of this socioeconomic adversity. We assessed the potential link between ADI and children's intertemporal decision-making through delay discounting at a 1-year follow-up. PLEs, encompassing distress, delusional, and hallucinational symptoms, were evaluated at both 1-year and 2-year follow-ups. Our analysis spans multiple follow-up periods and PLE indicators to investigate the sustained influence of ADI over time and to explore differential associations with various PLE symptoms, particularly delusional versus hallucinational.

Results

The demographic characteristics of the final sample (N=2,135) are presented in **Table 1**. Within the sample, 46.14% were female, 76.63% of participants had married parents, the mean household income was \$116,538, and 65.57% identified their race/ethnicity as white.

In our initial exploratory analysis, partial correlations were used to examine the relationship between psychopathological symptoms and delay discounting. Among the symptoms assessed (e.g., depression, anxiety, ADHD), only PLEs showed significant correlation with delay discounting (Spearman $\rho = -0.067$, p-FDR = 0.024 $\sim \rho = -0.057$, p-FDR = 0.035) (**Table S1**). This finding underscores the unique association between PLEs and delay discounting, laying the groundwork for subsequent investigations into how delay discounting—along with its genetic and neural correlates—may be associated with the heterogeneous relationship of neighborhood socioeconomic adversity with PLEs.

Neighborhood Socioeconomic Adversity is Associated with Steeper Delay Discounting and Higher PLEs

IV Forest analyses suggested that a higher ADI has significant associations with a steeper delay discounting ($\beta = -1.73$, p-FDR = 0.048) and a higher PLEs (distress score 1-year follow-up: $\beta = 1.872$, p-FDR = 0.048; distress score 2-year follow-up: $\beta = 1.504$, p-FDR = 0.039; delusional score 1-year follow-up: $\beta = 5.97$, p-FDR = 0.048; delusional score 2-year follow-up: $\beta = 4.022$, p-FDR = 0.048; hallucinational score 1-year follow-up: $\beta = 3.761$, p-FDR = 0.048; hallucinational score 2-year follow-up: $\beta = 4.786$, p-FDR = 0.039) (**Table 2**).

To evaluate the robustness of our findings from the IV Forest, we calculated the E-values for the average relationships from the IV Forest. The E-values quantify the minimum strength of association that unobserved confounders would need to possess with both ADI and the outcomes, conditional on the observed covariates in the IV Forest model, to nullify the observed relationships (68). E-values indicated that unobserved confounders would need to have a relative risk greater than 9.13 for delay discounting and between 7.32 and 457.03 for PLEs to account entirely for the observed effects (**Table S2**). This suggests a high threshold for unobserved confounding effects, thereby strengthening the validity of our analyses.

Supplementary analyses employing a conventional linear IV regression (60) and an alternative causal machine learning method, i.e., *Double ML* (69, 70), corroborated these findings. The conventional IV regression also showed that ADI has negative relationship with childhood delay discounting ($\beta = -0.468$, $p\text{-FDR} = 0.03$) and positive PLEs (distress score 1-year follow-up: $\beta = 0.609$, $p\text{-FDR} = 0.011$; distress score 2-year follow-up: $\beta = 0.78$, $p\text{-FDR} = 0.003$; delusional score 1-year follow-up: $\beta = 0.486$, $p\text{-FDR} = 0.028$; delusional score 2-year follow-up: $\beta = 0.578$, $p\text{-FDR} = 0.013$; hallucinational score 1-year follow-up: $\beta = 0.604$, $p\text{-FDR} = 0.011$; hallucinational score 2-year follow-up: $\beta = 0.827$, $p\text{-FDR} = 0.003$). The partial-linear IV model of the Double ML algorithm showed significant associations of ADI with children's delay discounting ($\beta = -0.429$, $p\text{-FDR} = 0.044$), distress score PLEs (1-year follow-up: $\beta = 0.495$, $p\text{-FDR} = 0.023$; 2-year follow-up: $\beta = 0.609$, $p\text{-FDR} = 0.005$), hallucinational score PLEs (1-year follow-up: $\beta = 0.498$, $p\text{-FDR} = 0.018$; 2-year follow-up: $\beta = 0.683$, $p\text{-FDR} = 0.002$), and 2-year follow-up delusional score PLEs ($\beta = 0.417$, $p\text{-FDR} = 0.044$). The negative relationship of ADI with 1-year follow-up delusional score PLEs were marginally significant ($\beta = 0.393$, $p\text{-FDR} = 0.051$). These results of the conventional linear IV regression (**Table S3**) and Double ML partial-linear IV regression (**Table S4**) are consistent with the findings obtained from the

IV Forest, further supporting the primary analyses and conclusions drawn from the study.

Genetic and Neural Correlates of Delay Discounting Moderates Vulnerability to Neighborhood Socioeconomic Adversity

Next, we moved beyond the average relationships to test for heterogeneous associations of neighborhood socioeconomic deprivation: i.e., whether the associations of ADI with PLEs varies systematically across children, and, if so, whether the heterogeneity is linked to individual's neurodevelopmental characteristics and the relevant genetic factors correlated to intertemporal valuation. Conceptually, this is analogous to testing for an interaction, where the associations of ADI with PLEs is moderated by, or differ as a function of a child's individual genetic and neural markers. Unlike traditional models might test a few pre-specified interactions, our IV Forest approach allows us to explore this heterogeneity in a data-driven manner across a high-dimensional set of potential moderators, including GPS and structural MRI and MID task functional MRI data.

To identify the best subset of genetic and neural correlates of delay discounting, we first selected GPS and MRI brain regions of interest (ROIs) specifically related to delay discounting. To analyze the nonparametric correlations of multiple input variables, we used a random forest-based feature selection *Boruta* algorithm (71). Its robustness and effectiveness in selecting relevant features in high dimensional, intercorrelated biomedical data (e.g., MRI) has been validated (71) and consistently applied in genetics and neuroscience research (72-74). The variables significantly correlated with delay discounting (p -Bonferroni <0.05) were GPS of cognitive performance, IQ, and education attainment; morphometric features (e.g., surface area, volume) in the limbic system (temporal pole, parahippocampal gyrus, caudate nucleus, rostral

285 anterior cingulate, isthmus cingulate), inferior frontal gyrus (pars opercularis), and fusiform
286 gyrus; mean beta activations of rewards/losses versus neutral feedback in the subcortical areas
287 (thalamus proper, ventral diencephalon), precentral gyrus, supramarginal gyrus, temporal lobe
288 (transverse temporal gyrus, superior temporal gyrus), and insula (**Table S5**).

289 We then assessed the heterogeneous associations of ADI with PLEs using three distinct IV
290 Forest models: (i) the Delay Discounting model, incorporating sociodemographic features and
291 delay discounting; (ii) the Gene-Brain model, which included sociodemographic features and
292 genetic and neural correlates of delay discounting (i.e., GPS and brain ROIs identified using
293 the Boruta algorithm); and (iii) the Integrated model which combined all the variables from the
294 previous two models (**Fig. 1**). All three models satisfied the overlap assumption (i.e., the
295 estimated propensity scores are not close to one or zero), which is crucial for the validity of the
296 estimated heterogeneous relationships (**Fig. S1**). In line with prior studies (75, 76), we obtained
297 conditional average effects, divided subjects into deciles (Q1: most vulnerable; Q10: most
298 resilient) based on the conditional average effects, and conducted three hypothesis tests (77)
299 on each model to determine the most effective model for capturing the individual differences
300 (heterogeneity) in the associations of ADI with PLEs: monotonicity, alternative hypothesis, and
301 ANOVA.

302 Among the three models, only the Integrated model showed significant individual
303 differences in the associations of ADI with PLEs. This was evident in the associations of ADI
304 with 1-year follow-up distress score PLEs (monotonicity test: $p\text{-FDR}=0.011$; alternative
305 hypothesis test: $p\text{-FDR}=0.0252$; ANOVA test: $p\text{-FDR}<0.001$) and 1-year follow-up
306 hallucinational score PLEs (monotonicity test: $p\text{-FDR}=0.038$; alternative hypothesis test: $p\text{-}$
307 $\text{FDR}=0.0252$; ANOVA test: $p\text{-FDR}<0.001$), as presented in **Fig. 3 and Table S6**. In contrast,
308 the Delay Discounting model and Gene-Brain model failed to pass the heterogeneity tests

(monotonicity test: $p\text{-FDR} \geq 0.05$; alternative hypothesis and ANOVA test: $p \geq 0.05$).

To elucidate the role of specific genetic and neural correlates within the heterogeneous associations of ADI with PLEs, we obtained Shapley additive explanation (SHAP) scores (78). SHAP scores provide insights into how each variable correlates positively or negatively to the differential associations of ADI with 1-year follow-up observations of distress score and hallucinational score PLEs. These scores help differentiate the roles of these factors across subgroups, ranging from low to high conditional average effects, thereby providing a nuanced understanding of how ADI is correlated with PLEs through various genetic and neural pathways.

In both distress score and hallucinational score PLEs, children who showed higher levels of ADI's negative relationship with PLEs exhibited distinct neuroanatomical and functional brain patterns, particularly in the limbic system. These patterns included reduced neuroanatomical features such as smaller white matter and surface area in the right temporal pole, reduced area and volume in the right parahippocampal region, decreased left white surface area, smaller area in the right isthmus cingulate, reduced intracranial volume, smaller caudate nucleus volume, and lower total grey matter volume. Functionally, greater activation during MID tasks was observed in several areas including the posterior cingulate, right ventral diencephalon, right insula, left thalamus proper, and left precentral gyrus. Additionally, children more adversely affected by ADI, as indicated by higher conditional average effects on PLEs, exhibited larger right fusiform volume, decreased activation in the left superior temporal gyrus, younger parental age, and lower BMI (**Fig. S2**).

The analysis also revealed that higher conditional average effects on distress score PLEs was associated with higher cognitive performance GPS and a lower likelihood of being

Hispanic. In contrast, for hallucinational score PLEs, greater importance was attributed to increased activation in the left supramarginal gyrus during MID tasks and more pronounced discounting of future rewards. These nuanced associations are depicted in **Fig. S2**.

To further characterize this gene-environment interaction underlying these relationships, we obtained conditional average effects of ADI on 1-year follow-up distress score PLEs, conditioned on GPS for cognitive traits (i.e., cognitive performance, IQ, and education attainment). Within the model, the estimated conditional average effects tended to increase with higher values of GPSs (**Fig. 4**). This suggests that individuals with a higher genetic predisposition for academic and cognitive achievement are paradoxically more vulnerable to the psychotogenic effects of a deprived socioeconomic environment.

Given that quality control procedures reduced a large proportion of participants from the initial ABCD Study cohort, we conducted sensitivity analysis using inverse probability weighting to adjust for potential selection bias (see **Supplementary Information** for methodological details). After inverse probability weighting adjustments, the heterogeneous associations of ADI remained statistically significant for both distress (monotonicity: $p\text{-FDR}=0.011$; alternative hypothesis: $p\text{-FDR}=0.008$; ANOVA: $p\text{-FDR}<0.001$) and hallucinational PLEs (monotonicity: $p\text{-FDR}=0.038$; alternative hypothesis: $p\text{-FDR}=0.008$; ANOVA: $p\text{-FDR}<0.001$) at 1-year follow-up (**Table S8**).

Discussion

In this study, we examined how neighborhood socioeconomic deprivation relates to children's intertemporal choice behavior (delay discounting) and PLEs, considering the multifaceted associations of neighborhood adversity and its underlying biological, environmental, and behavioral drivers. Our findings can be distilled into two main points. Firstly, there was a notable link of living in socioeconomically disadvantaged neighborhoods to the propensity for children to prefer immediate rewards over larger, delayed ones—a behavior known as steep delay discounting (indicative of lower impulse control) and to a higher rate of PLEs. This association was significant even after adjusting for a range of observed confounding factors (e.g., familial socioeconomic status) and reducing potential bias from unobserved confounders via IV approach. Secondly, the associations of disadvantaged neighborhood environments with PLEs were found to be heterogeneous. This individual variability is moderated not just by delay discounting, but also by a confluence of factors including genetic predisposition for cognitive intelligence, and brain morphometry and functioning (task activation). Causal machine learning models utilized in our study have identified a spectrum of conditions that either exacerbate vulnerability or contribute to resilience, accounting for the diverse correlations of neighborhood environments with children's PLEs.

Our findings hold implications for social science and economic theory by providing quasi-experimental evidence for the “behavioral poverty trap”. Our finding that neighborhood socioeconomic adversity is linked to children's intertemporal choice toward steeper reward discounting challenges the longstanding economic assumption that time preference is a fixed trait. This environmentally induced shift towards more impulsive decision-making, likely

driven by factors such as cognitive load from resource scarcity and reduced social trust, could foster behaviors that might perpetuate socioeconomic challenges. These results suggest that policies aimed at improving childhood environments may be crucial for breaking intergenerational cycles of disadvantage. A more detailed examination of these theoretical frameworks is provided in the **Supplementary Information**.

A plausible biological mechanism for these findings involves the association of chronic stress with the neurodevelopmental trajectory of prefrontal-limbic circuits. Prior research indicates that neighborhood socioeconomic deprivation is a potent chronic stressor that elevates glucocorticoid hormones like cortisol (79-83). During the critical adolescent period of our cohort, prolonged glucocorticoid dysregulation can induce epigenetic modifications that alter the mesocorticolimbic dopaminergic system, which governs reward valuation and is heavily implicated in the pathophysiology of psychosis (33, 84-87). The neuro-genetic patterns of vulnerability we identified—characterized by reduced limbic volume and altered prefrontal-limbic reactivity during reward processing—are therefore consistent with a stress-induced disruption of the reward system that confers risk for both impulsive decision-making and PLEs (24, 88). A more detailed review of the supporting evidence is available in the **Supplementary Information**.

In our study, we discovered that children, when exposed to deprived neighborhoods and already facing the challenges of residential disadvantage, were more likely to experience PLEs. Surprisingly, these children also showed a higher GPS for cognitive performance, IQ, and education attainment (**Fig. 4**). At first glance, this finding seems to contradict prior research, which has consistently identified a negative relationship between PLEs and cognitive performance (38, 89).

We interpret this paradox through the lens of gene-environment interaction theory, specifically the bioecological model and the Scarr-Rowe hypothesis (90-92). This framework posits that genetic influences on traits diminish under adverse environmental conditions— analogous to how nutrient-poor soil constrains plant growth regardless of genetic potential (93). In our context, residential disadvantage appears to suppress the protective effects of cognitive-related genetic variants, rendering children with higher cognitive ability GPS more vulnerable to environmental stressors that precipitate PLEs. Critically, those with the greatest genetic potential for cognitive performance lose the most protection, creating heightened susceptibility to neighborhood-level adversity.

This interaction pattern aligns with differential susceptibility theory (94, 95), which proposes that certain genetic profiles confer both vulnerability to negative environments and enhanced benefit from positive ones. While our study of neighborhood deprivation captures only the “for worse” component, our results satisfy a key theoretical prerequisite by providing evidence that genetic advantage may paradoxically increase environmental vulnerability. This challenges traditional diathesis-stress models and instead supports a genetic plasticity framework where the same variants that confer advantage in supportive contexts create risk in adverse ones.

The three GPSs examined here—encompassing cognitive performance, IQ, and educational attainment—capture accumulated learning and developmental potential (96, 97). Within the differential susceptibility framework, elevated cognitive ability GPS may index heightened neurobiological plasticity that, when confronted with resource-poor environments, generates stress responses that increase PLE risk. Conversely, individuals with lower cognitive ability GPS may show resilience precisely because they exhibit reduced environmental reactivity, making them less sensitive to the specific stressors—such as thwarted potential or

421 limited opportunities—that characterize deprived neighborhoods and may trigger psychotic-
422 like experiences.

423 Consistent with our findings, recent large-scale studies have demonstrated that the
424 association of genetics with brain structure, cognitive functions, and mental health disorders
425 becomes less significant in harmful environments (e.g., abuse) (98, 99). Conversely, in more
426 supportive and enriched settings, like those associated with higher socioeconomic status,
427 genetic influences are more noticeable (e.g., high socioeconomic status) (91, 100, 101).
428 Together with these findings, our study contributes to a deeper understanding of how genetic
429 and environmental factors interact to influence the development of psychopathology in children.

430 Several limitations of this study warrant consideration. First, while our IV approach was
431 employed to mitigate unobserved confounding, the observational nature of the ABCD Study
432 necessitates a cautious interpretation of causality. While we took steps to substantiate it with
433 prior research and sensitivity analyses (detailed in *Instrumental Variable Regression* section of
434 the **Supplementary Information**), our interpretations should be read as associations based on
435 IV-estimates that are compatible with a causal account but remain assumption-dependent.

436 Second, the stringent neuroimaging and behavioral quality-control filters necessary for
437 internal validity yielded appreciable attrition from the ABCD Study cohort, introducing
438 potential selection effects. The final sample was predominantly white (65.57%, self-reported
439 race/ethnicity), which may also limit the generalizability of our specific neuro-genetic findings
440 to individuals from other racial and ethnic backgrounds. To formally address this challenge, we
441 conducted a sensitivity analysis using inverse probability weighting, a rigorous method
442 designed to account for potential selection bias. While this sensitivity analysis demonstrated
443 that our key findings on heterogeneous associations of neighborhood adversity with PLEs were

robust to reweighting on observed characteristics, this approach cannot address selection on unmeasured factors. Children excluded due to motion artifacts or task non-compliance may differ in ways correlated with both neuroimaging quality and psychiatric vulnerability. Therefore, our findings apply specifically to children capable of completing comprehensive assessments—a limitation that may exclude those at highest risk. Future studies should develop protocols enabling broader participation while maintaining data quality.

Third, the follow-up horizon is limited to baseline, 1-year, and 2-year assessments. It should be noted that additional longitudinal data from the ABCD Study were publicly released after this work was finalized, so were not included. Longer observation windows are needed to map trajectories from adversity to psychosis-spectrum outcomes and to evaluate the stability of heterogeneity signals.

Beyond neighborhood deprivation, testing other environmental risk factors as primary exposures—particularly parenting/ family environment (102) and early-life stress (99)—will clarify whether similar neuro-genetic profiles index differential susceptibility across contexts. Later ABCD releases enable (i) time-varying and cumulative-exposure models, (ii) ancestry-stratified and site-robust analyses, and (iii) external replication. Such work will help determine whether the identified moderators can support risk stratification and inform precision prevention.

This study highlights the differential associations of neighborhood disadvantage with intertemporal economic decisions and PLEs during early childhood. It underscores the importance of identifying diverse links by integrating genetic and environmental factors to guide personalized healthcare approaches. Furthermore, we propose that enhancing the childhood environment may contribute to the reduction of economic and health inequality gaps.

Economic policies promoting positive intertemporal choice (e.g., increased savings, healthy diet) have predominantly focused on paternalistic welfare policies in adulthood. These policies often assume that an individual's tendency to discount future rewards is fixed ("exogenous") (24). However, our findings suggest that policies or interventions aimed at enhancing the socioeconomic environment during childhood may foster improved intertemporal choice behavior, thereby reducing economic (13, 20, 21) and health inequality (17, 103). By addressing the root of the problem, this indirect approach may assist individuals in developing the capacity to make more informed choices, ultimately promoting better outcomes.

Finally, our findings provide an empirical counterpoint to longstanding philosophical questions about human agency. Immanuel Kant, in his book *Critique of Practical Reason*, argued for an a priori capacity for reason, independent of external factors (104). Our work, however, suggests that the neuro-genetic mechanisms governing intertemporal choice and psychiatric vulnerability are profoundly associated with a child's socioeconomic environment. This link between environment and the biological substrates of decision-making challenges models of responsibility that assume unconstrained free will. Instead, our results call for a more nuanced understanding of individual agency, suggesting that societal approaches to justice, policy, and intervention must account for the powerful and heterogeneous ways in which a child's environment is related with their neurocognitive development.

Data availability

The ABCD Study dataset is openly available to all eligible researchers upon the submission of an access request via the National Institutes of Mental Health Data Archive (<https://nda.nih.gov/abcd>). Comprehensive written informed consent was obtained from the parents of participants, with children providing assent. The study protocols were approved by the University of California, San Diego's Institutional Review Board (IRB), under approval number 160091, in addition to receiving approval from the IRBs of the 21 participating data collection sites (105).

Code availability

All codes needed to replicate the results can be found at <https://github.com/Transconnectome/DD-HTE>.

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981 **Tables and Figures**

982 **Table 1. Socioeconomic/demographic characteristics of the participants.** *Age* is rounded to chronological month. *Race/Ethnicity* denote child’s self-
983 reported racial / ethnic identity. *Household Income* is assessed as the total combined family income for the past 12 months. *Parental Education* is measured
984 as the highest grade or level of school completed or highest degree received. *Family History of Psychiatric Disorders* represents the proportion of first-degree
985 relatives who experienced mental illness.

Demographic Characteristics		N	Ratio (%)	Mean (SD)
Age		2,135		120.1541 (7.4658)
Sex	Male	1,150	53.86%	
	Female	985	46.14%	
Marital Status of the first caregiver	Married	1,636	76.63%	
	Widowed	12	0.56%	
	Divorced	193	9.04%	
	Separated	62	2.9%	
	Never Married	142	6.65%	
	Living with Partner	90	4.22%	
Race/Ethnicity	White	1,400	65.57%	
	Black	136	6.37%	
	Hispanic	373	17.47%	

	Asian	7	0.33%	
	Other	219	10.26%	
Parent's Identity	Biological Mother	1,848	86.56%	
	Biological Father	215	10.07%	
	Adoptive Parent	39	1.83%	
	Custodial Parent	12	0.56%	
	Other	21	0.98%	
Household Income		2,135		\$116,538 (70,669)
Parental Education		2,135		17.2838 (2.3046)
BMI		2,135		18.4298 (3.8572)
Parental Age		2,135		40.8775 (6.3825)
Family History of Psychiatric Disorders		2,135		0.0958 (0.1125)

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991 **Table 2. Potential associations of neighborhood socioeconomic adversity with intertemporal valuation and PLEs.** Average relationships of ADI with
 992 delay discounting and PLEs in the IV Forest models are shown. All p-values were corrected for multiple comparison using false discovery rate.

	IV Forests: Average Associations				
	Estimates	Std. Error	95% Lower CI	95% Upper CI	P-FDR
Delay Discounting (1-year follow-up)	-1.730	0.748	-3.195	-0.265	0.048
Distress Score PLEs (1-year follow-up)	1.872	0.612	0.673	3.071	0.048
Distress Score PLEs (2-year follow-up)	1.504	0.592	0.345	2.664	0.039
Delusional Score PLEs (1-year follow-up)	5.970	2.911	0.264	11.676	0.048
Delusional Score PLEs (2-year follow-up)	4.022	1.987	0.127	7.917	0.048
Hallucinational Score PLEs (1-year follow-up)	3.761	1.902	0.033	7.489	0.048
Hallucinational Score PLEs (2-year follow-up)	4.786	1.861	1.139	8.433	0.039

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Fig. 1. Study flow diagram. This figure illustrates the participant selection and data processing in our study. We initially included 11,876 participants aged 9-12 years from the Adolescent Brain Cognitive Development (ABCD) Study, utilizing the release 4.0 dataset which encompasses baseline, 1-year follow-up, and 2-year follow-up observations. We excluded observations not meeting the ABCD Study's MRI quality control standards and those failing the Johnson-Bickel validation criterion for delay discounting. Participants with missing values in genotypes, MRI, NIH Toolbox Cognitive Battery, delay discounting, residential address, ADI, and PLEs were also removed. Subsequently, sociodemographic features underwent kNN imputation. This resulted in a final sample size of N=2,135. Using this cohort, our study first investigated the average associations of neighborhood socioeconomic deprivation with children's intertemporal valuation of rewards and PLEs. We then explored the individual differences of these links in relation to children's delay discounting behaviors and associated genetic and neural factors.

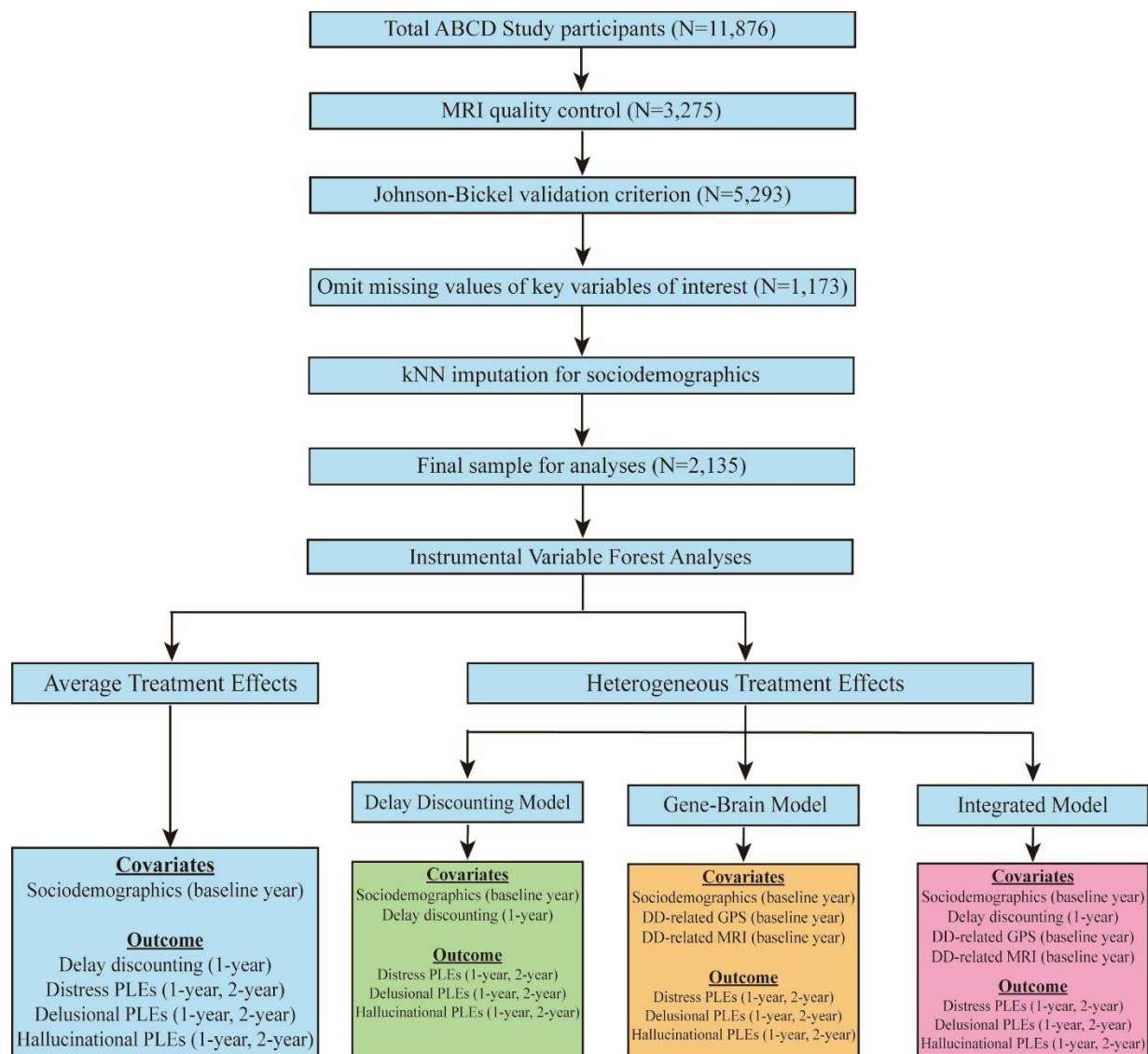
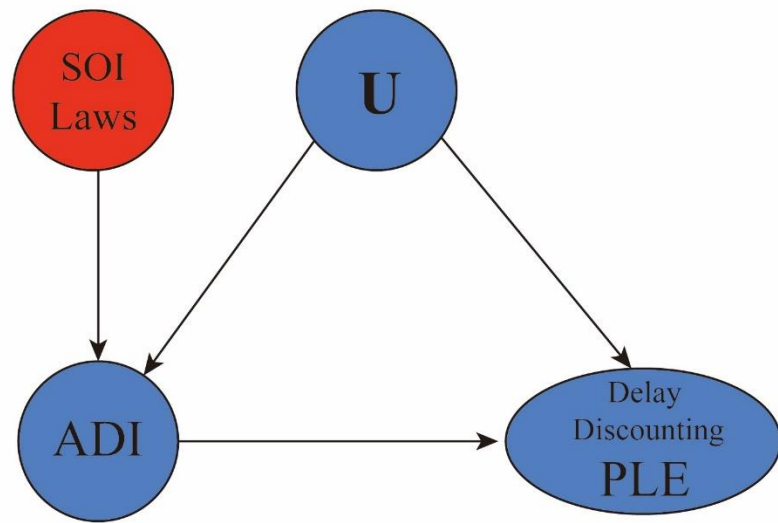
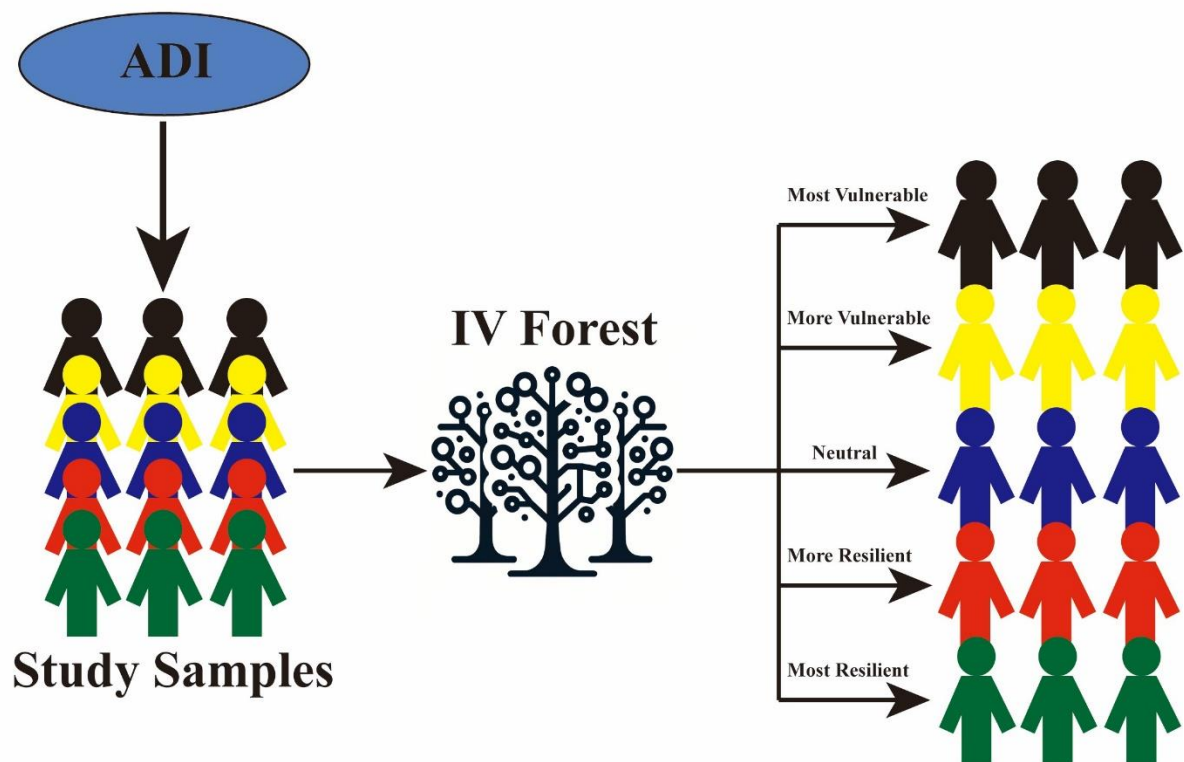


Fig. 2. Illustration of instrumental variables analysis. (A) Graphical model of the IV method. This panel depicts how the IV method is applied to address the potential biases from unobserved confounders U when analyzing the effects of exposure/treatment (ADI) on outcomes (delay discounting and PLEs) within the observational context of the ABCD Study. Using the instrument (SOI laws), the IV method aims to partial out the influence of U on ADI, thus adjusting for potential confounding biases. (B) Graphical representation of the IV Forest algorithm. The IV Forest algorithm stratifies subjects in a manner that maximizes the individual differences in predicted conditional average effects. This panel illustrates the use of IV Forest to assess the association of ADI with delay discounting and PLEs, as well as to explore the heterogeneous relationships of ADI with PLEs, factoring in individual variations in delay discounting and its associated genetic and neural correlates.

A.



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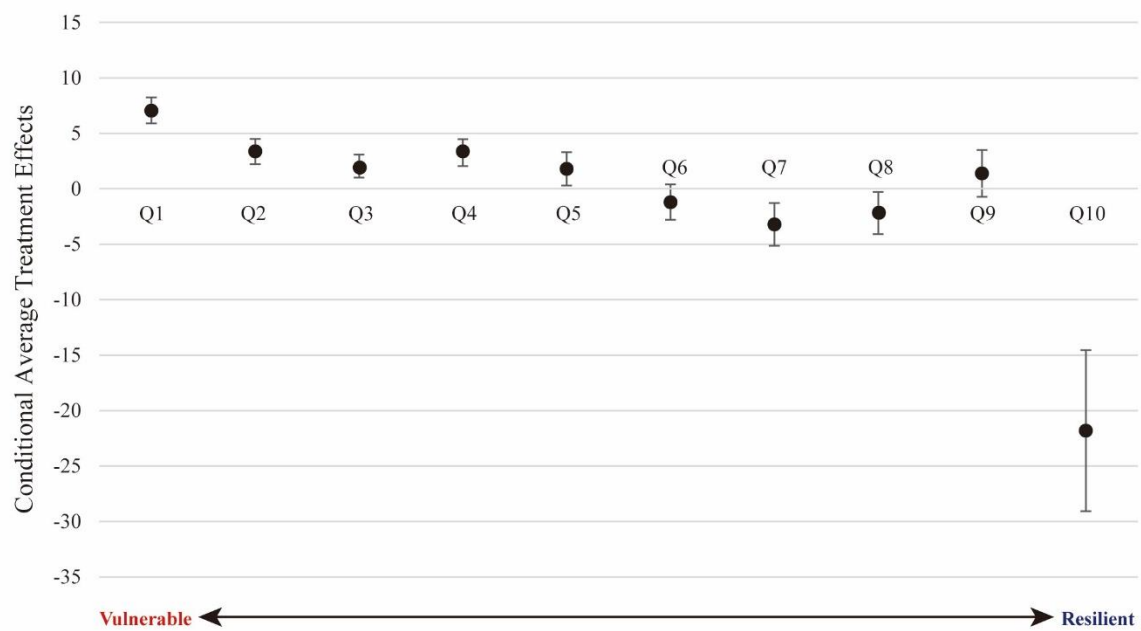
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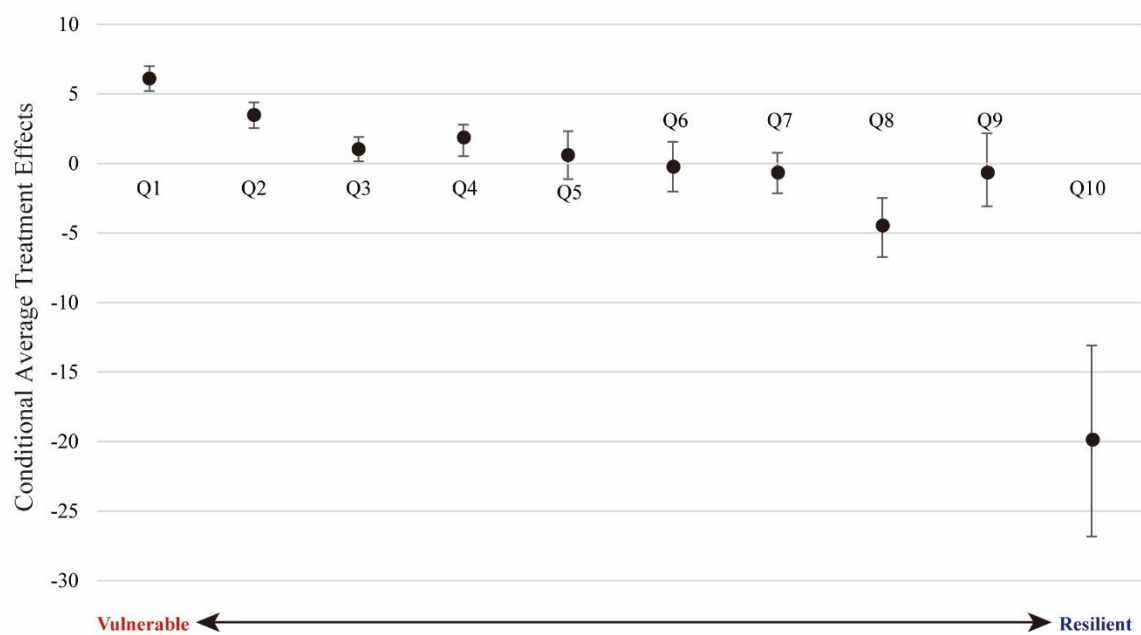
Fig. 3. Delineation of heterogeneous associations by vulnerable/resilient subgroups.

Heterogeneity in the associations of ADI with PLEs are shown as a bar plot, specifically focusing on 1-year follow-up distress score PLEs (**A**) and hallucinational score PLEs (**B**). These relationships are plotted across ten deciles, which are organized based on the relative vulnerability or resilience of the participants, with Q1 denoting the most vulnerable and Q10 indicating the most resilient. Point estimates of the conditional average effect of each decile were derived via a doubly-robust estimation method within the IV Forest algorithm. 95% confidence intervals are depicted using error bars.

A. Distress Score PLEs 1-year follow-up



B. Hallucinational Score PLEs 1-year follow-up



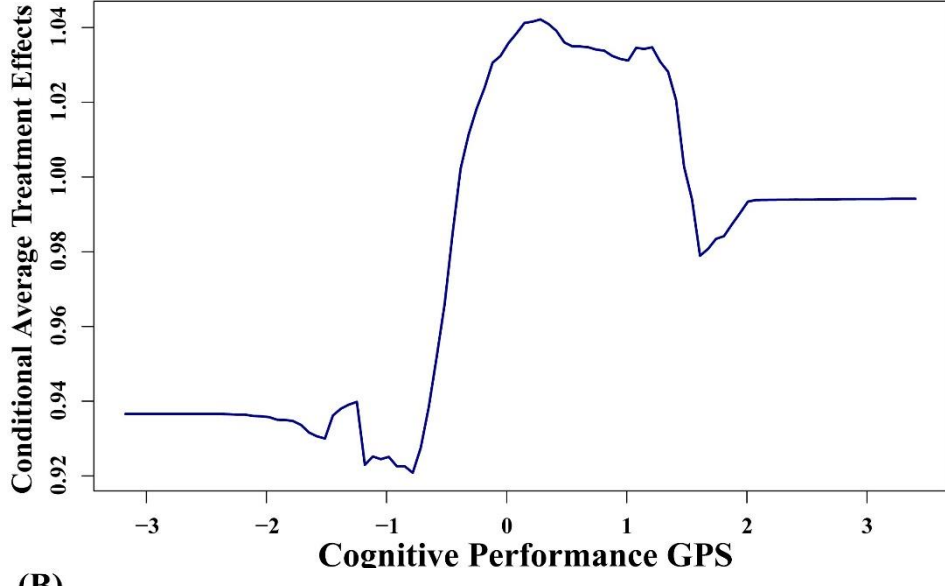
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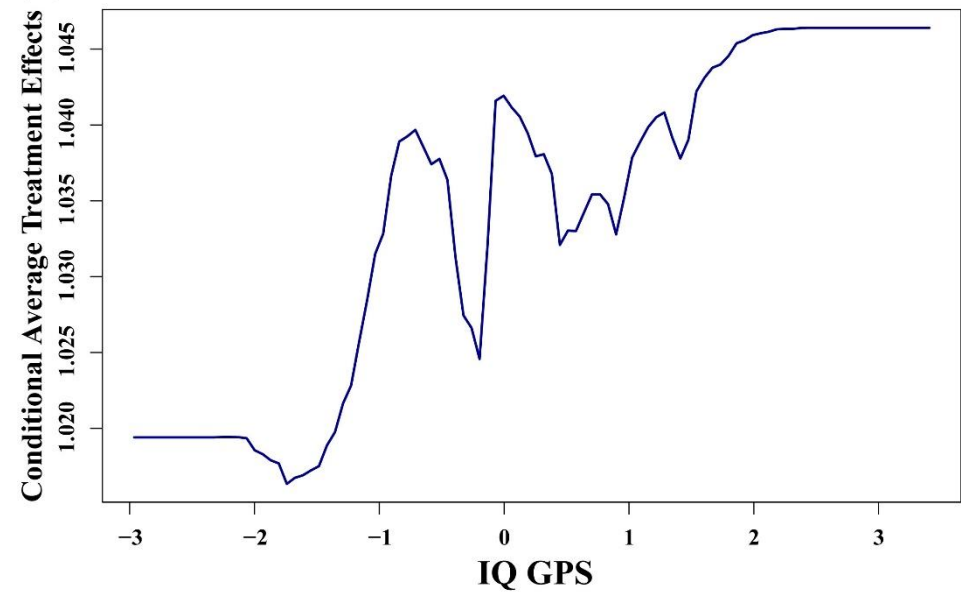
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Fig. 4. GPS for cognitive traits moderate environmental risk for PLEs. Plotted is the conditional average effects of neighborhood socioeconomic deprivation (ADI) on distress score PLEs at 1-year follow-up. The effect of ADI is shown as a function of an individual's genetic predisposition (GPS) for (A) cognitive performance, (B) IQ, and (C) education attainment, revealing complex patterns of gene-environment interaction.

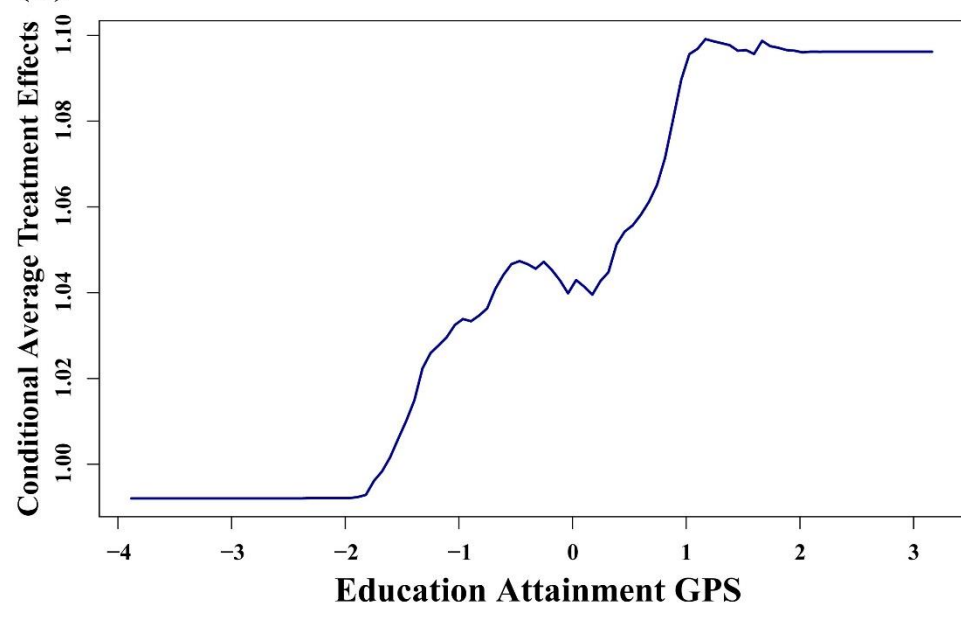
(A)



(B)



(C)



Supplementary Information for

Quasi-Experimental Analysis Reveals Neuro-Genetic Susceptibility to Neighborhood Socioeconomic Adversity in Children's Psychotic-Like Experiences

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Supplementary Information

Data

Neighborhood Disadvantage

Neighborhood disadvantage was measured with Residential History Derived Scores based on the Census tracts of each respondent's primary addresses by the ABCD team. Consistent with prior research (2, 45), we chose national percentile scores of the Area Deprivation Index (ADI) in baseline year, calculated from the 2011~2015 American Community Survey 5-year summary. It has 17 sub-scores regarding various socioeconomic factors such as median household income, income disparity, percentage of population aged more than 25 years or more with at least a high school diploma, and percentage of single-parent households with children aged less than 18 years, etc. Higher values of the ADI indicate greater neighborhood disadvantage.

Delay Discounting

Delay discounting was measured by the adjusting delay discounting task in the 1-year follow-up ABCD data (51, 106). Each child was asked to make choices between a small immediate hypothetical reward or a larger hypothetical \$100 delayed reward at multiple future time points (6h, one day, one week, one month, three months, one year, and five years). By increasing or decreasing the smaller immediate reward depending on the child's response, the task records the indifference point (i.e., the small immediate amount deemed to have the same subjective value as the \$100 delayed reward) at each of the seven delay intervals. Test-retest reliability of this delay discounting measure has been validated (107, 108). Studies show that preadolescent children are capable of comprehending the delay discounting task and show similar patterns of discounting as adults (109).

To avoid methodological problems regarding mathematical discounting models (hyperbolic vs. exponential) and positively skewed parameters of discounting functions (108, 110), we used the area under the curve, a model-free measure of delay discounting (110). The area under the curve measure of delay discounting ranges from 0 to 1, with lower values indicating steeper discounting and higher impulsivity.

Psychotic-Like Experiences

First and second-year follow-up observations of psychotic-like experiences (PLEs) were measured using the Prodromal Questionnaire-Brief Child Version (PQ-BC; child-reported). PQ-BC has a 21-item scale validated for use with a non-clinical population of children aged 9-10 years (53, 111). In line with the previous research (2, 53, 54, 111), we computed *Distress Score*, indicating the levels of total distress from psychotic-like symptoms. Distress Score is the weighted sum of responses with the levels of distress, ranging from 0 to 126. Additionally, to test whether the heterogeneous associations of neighborhood adversity differ among psychotic symptoms, Distress Score was divided into two separate scores: *Delusional Score* and *Hallucinational Score* (1, 112). A higher value indicates greater severity of PLEs.

Genome-wide Polygenic Scores

Children's genetic predispositions were assessed with genome-wide polygenic scores (GPS). Summary statistics from genome-wide association studies were used to generate GPS of cognitive intelligence (cognitive performance (96), education attainment (96), IQ (113)), psychiatric disorders (major depressive disorder (114), post-traumatic stress disorder (115), attention-deficit/hyperactivity disorder (116), obsessive-compulsive disorder (117), anxiety

(118), depression (119), bipolar disorder (120), autism spectrum disorder (121), schizophrenia (122), cross disorder (123)), and health and behavioral traits (BMI (124), neuroticism (125), worrying (125), risk tolerance (126), automobile speeding propensity (126), eating disorder (127), drinking (126), smoking (126), cannabis use (128), general happiness (129), snoring (130), insomnia (130), alcohol dependence (131)). PRS-CSx, a high-dimensional Bayesian regression framework that places continuous shrinkage prior on single nucleotide polymorphisms effect sizes (132), was applied to enhance cross-population prediction. This method has consistently shown superior performance compared to other methods across a wide range of genetic architectures in simulation and real data analyses (132). Hyperparameter optimization for the GPSs was conducted using a held-out validation set of 1,579 unrelated participants. Adjustments for population stratification were performed based on the first ten ancestrally informative principal components to account for potential confounding effects.

Anatomical Brain Imaging: T1/T2, Freesurfer 6

Baseline year T1-weighted (T1w) 3D structural MRI acquired in the ABCD study were processed following established protocols (50, 133): To maximize geometric accuracy and image intensity reproducibility, gradient nonlinearity distortion was corrected (134). After correcting intensity nonuniformity using tissue segmentation and spatial smoothing, images were resampled to 1 mm isotropic voxels. We used Freesurfer v6.0 (<https://surfer.nmr.mgh.harvard.edu>) for the following procedures: cortical surface followed by skull-stripping (135), white matter segmentation, and mesh creation (136), correction of topological defects, surface optimization (137), and nonlinear registration to a spherical surface-based atlas (138, 139). Using Desikan–Killiany atlas (140), a standard atlas for Freesurfer and ABCD study, we extracted 399 brain ROI

measures, including volumes, surface area, thickness, mean curvature, sulcal depth, and gyrification.

Functional MRI (fMRI): Monetary Incentive Delay (MID) task

The MID task was used measure the neural activation during anticipation and receipt of monetary gains and losses. In each trial, participants were shown a graphical cue of the 5 possible incentive types: large reward (\$5), small reward (\$0.20), large loss (-\$5), small loss (-\$0.20), or neutral (\$0). The incentive cue is presented for 2,000 ms, followed by a jittered anticipatory delay (1,500–4,000 ms). Subsequently, a target to which participants respond to gain or avoid losing money was shown (150–500 ms), and feedback of their performance was provided (2,000 ms). A total of 40 reward, 40 loss, and 20 neutral trials were presented in pseudo-random order across the two task runs. Task parameters was dynamically manipulated for each subject to maintain 60% success rate (50). We used baseline year observations of average beta weights of the MID task fMRI with Desikan-Killiany parcellations (140).

Statistical Analyses

Instrumental Variable Regression

The IV method controls unobserved confounding bias by utilizing an instrumental variable Z which affects the exposure variable of interest X but has no direct effect on the outcome variable Y (60). To estimate the associations of neighborhood adversity (ADI) with our outcomes, we must address the endogeneity of ADI, which arises from its correlation with unobserved confounders. Hausman test (141) results confirmed this endogeneity (all $p \leq 0.0158$). This

underscores the necessity of employing IV regression approach to control for the significant confounding bias and to test the potential relationship of neighborhood disadvantage with delay discounting and PLEs.

In our study, the instrumental variable Z for the exposure of ADI was the presence of state-level source of income (SOI) laws at baseline assessment, which prohibit income discrimination in the housing market. SOI laws are designed to ensure that landlords cannot refuse housing vouchers, which are provided to low-income families to assist in securing quality housing. Such legislation is critical because, despite the intention behind vouchers, many landlords prefer direct cash payments and might otherwise decline voucher-based payments. Thus, families residing in states with SOI laws are more likely to have better residential environments (i.e., lower neighborhood socioeconomic adversity).

The IV method requires the instrument Z to satisfies two core assumptions. First, the instrument must be strongly correlated with the exposure variable X . F-statistic above ten is considered a strong instruments (142). The F-statistic for each model was $F = 34.031$ ($p < 0.0001$), suggesting that the instrument SOI laws is strongly associated with ADI.

Second, the instrument must affect the outcome Y through its effect on the exposure X . We assume this exclusion restriction: SOI laws affect outcomes only through ADI. While we recognize that this assumption cannot be directly tested from data and may be imperfect, its plausibility is typically drawn from prior research and theoretical underpinnings (60). Reports from the US Department of Housing and Urban Development indicate that SOI laws increase landlords' acceptance of housing vouchers by 20.2%p to 59.3%p (143). Research links SOI laws with significant reductions in neighborhood poverty (144) and improved health outcomes in children, including lower hospitalization rates, less impulsive consumption (145), and

substantially better mental health (146). Taken together, the presence of SOI laws may affect cognitive and psychiatric outcomes—particularly delay discounting and PLEs—solely by enhancing neighborhood environment, conditional on the individual sociodemographic characteristics such as household income, parental education, and race/ethnicity. This relationship supports the plausibility of the exclusion restriction, crucial for the validity of our IV method.

All continuous variables were standardized (z-scaled), and analyses were run using *ivreg* (147) in R version 4.1.2. For all analyses in our study, threshold for statistical significance was set at two-tailed $p < 0.05$, with multiple comparison correction based on false discovery rate.

Causal Machine Learning

IV Forest (*grf* R package version 2.2.1) (65, 66) is a novel causal machine learning approach extends from the conventional random forest framework (148) with recursive partitioning, subsampling, and random splitting to identify the average effects and its individual differences.

Initially, the IV Forest randomly splits the dataset into two independent subsets, S and T. Subset S is dedicated solely to the construction of individual trees within the forest, where each tree explores potential divisions—such as "Race/ethnicity = White"—to split S into groups S_1 and S_2 , based on the fulfillment of the specified criteria. The selection of these splits is strategically chosen to maximize the differences in conditional average effects estimates between the groups S_1 and S_2 . Following the construction phase, subset T, independent of S, is employed for model validation. Fresh observations from T are introduced into groups with similar treatment responses by each tree. The aggregation of results from multiple trees is conducted through local weighting method, aimed at reducing overall estimate variance and improving accuracy. Using

1247 separate subsets for tree building and model validation ensures an honest estimation of
1248 conditional average effects, systematically reducing the risk of overfitting (67).

1249 In all IV Forest analyses, we accounted for the multi-site structure of the ABCD Study by
1250 incorporating study sites as a clustering variable. This approach addresses potential site-level
1251 confounding and ensures valid statistical inference in the presence of clustered data. Specifically,
1252 we implemented cluster-based subsampling: rather than randomly selecting individual
1253 participants to construct each tree, we sampled entire site clusters as the unit of analysis. This
1254 modification preserves the hierarchical data structure and prevents within-cluster correlation
1255 from biasing our estimates. Crucially, this approach maintains the honesty property of IV Forest
1256 by ensuring complete separation between training and inference samples. By using data from
1257 distinct clusters for model training versus prediction, we reduce the risk of information leakage
1258 that could otherwise inflate performance metrics and compromise the validity of our estimates.

1259 Using IV Forests, we obtained augmented inverse propensity weighted estimates of average
1260 effects, a doubly-robust estimator which can capture complex patterns of individual differences
1261 and reduce reliance on parametric model forms (65) such as linearity. This is particularly
1262 advantageous when the relationship between environmental variables and neurocognitive
1263 development is likely nonlinear (6, 149, 150).

1264 In line with prior studies (75, 76), we evaluated heterogeneous effects by testing whether the
1265 average effects are significantly different among subgroups defined by their relative
1266 resilience/vulnerability (77). These subgroups were defined across a decile spectrum, with Q1
1267 representing the most vulnerable and Q10 the most resilient. We considered a model to have
1268 significant heterogeneous effects only if it satisfied all three of the following criteria:

- 1269 (i) The monotonicity test evaluates the existence of at least one inequality in the average

effects across the deciles. This is achieved by whether to accept or reject the null hypothesis ($\mathcal{H}_0$), which states that effects are equal across all deciles. Essentially, the test determines whether there is a consistent, ordered relationship in the effects from one decile to the next, indicating a monotonic trend.

(ii) The alternative hypothesis test evaluates whether the average effect in the highest decile exceeds the combined average effects in the remaining deciles Q2 through Q10.

(iii) The ANOVA test determines whether the average effects are statistically different across deciles. In this context, the group mean in the ANOVA corresponds to the average effect of each decile.

To ensure that the IV Forest estimations are robust across different random seeds, we developed 100-seed ensemble IV Forest model. Specifically, we used the following procedures:

- 1) For each iteration, randomly split the data in half (i.e., train vs test sets) to build a forest model with the first half and perform estimation with the other half. We repeated this process 100 times using different seeds in each iteration to build 100 forest models.
- 2) Combine the 100 forest models into one big IV Forest model and then rank the observations into deciles according to their estimated conditional average effects.
- 3) Obtain augmented inverse propensity weighted average effects for each decile and perform monotonicity, alternative hypothesis, and ANOVA tests.

Having established significant heterogeneous associations of ADI with PLEs in our Integrated model, and identified cognitive performance GPS as one of the key moderators through SHAP analysis, we systematically examined gene-environment interactions across three

cognitive ability GPSs (i.e., cognitive performance, IQ, and education attainment).

To isolate the moderating role of each GPS using conditional effects, we employed a standardized procedure (151):

I. To isolate the effect of a single GPS variable, we constructed a synthetic dataset where its values spanned their full empirical range. All other covariates in the model were held constant at their respective sample means to create a consistent, average profile.

II. We then used our IV Forest model to predict the conditional average effects. This yielded an estimate of the link between ADI and PLEs for each point along the continuum of the specific GPS.

The resulting plot illustrates how the relationship of neighborhood socioeconomic deprivation with PLEs varies across the distribution of a specific GPS variable, holding all other factors constant.

Methodological Strengths of Causal Machine Learning

In this study, we utilized innovative causal machine learning techniques to test the negative relationship of neighborhood deprivation with childhood psychopathology. Specifically, we employed the IV Forest method that allows us to discern how residential deprivation relates to children's PLEs in a manner dependent on a variety of genetic risk factors (e.g., GPS of cognitive performance, educational attainment, and IQ (54, 89, 152)) and environmental risk factors (e.g., family income (2, 153)), as identified in existing literature. Our findings were adjusted to account for potential biases from both observed and unobserved variables.

The machine learning algorithm we used was adept at modelling the complex interplay gene-

environment interactions. Among the three IV Forest models we tested (i.e., Delay Discounting, Gene-Brain, Integrated), only the Integrated model—which included delay discounting, sociodemographic characteristics, and genetic and neural correlates of delay discounting—identified the significant heterogeneous relationships of ADI with children’s PLEs. This suggests that the intricate interactions among environmental, genetic, neural factors, and delay discounting play a crucial role in how socioeconomic adversity relates to PLEs.

The IV Forest model represents a significant advancement over traditional analysis methods by enabling data-driven feature selection and the stratification of heterogeneous associations (65, 66). Unlike methods that rely on patterns predetermined by researchers, the IV Forest model inductively identifies complex and nonlinear interactions, providing a deeper and more nuanced understanding of the data. Traditional deductive approaches often suffer from low statistical power and bias (59, 154), which inadequately capture the complexity of gene-environment interactions (42, 43). For instance, employing conventional linear regression to model interactions among the 45 covariates in our Integrated model would necessitate the inclusion of over 35 trillion interaction terms. This is not only impractical due to its complexity but also prone to issues like reduced statistical power, poor interpretability, and collinearity.

Given these challenges, we believe that analysis methods that assess heterogeneous associations based on machine learning hold significant potential as powerful tools for advancing precision science in psychology and medicine. These approaches provide a more dynamic and accurate framework for understanding the multifaceted influences on psychopathology, demonstrating significant promise for future research in these fields.

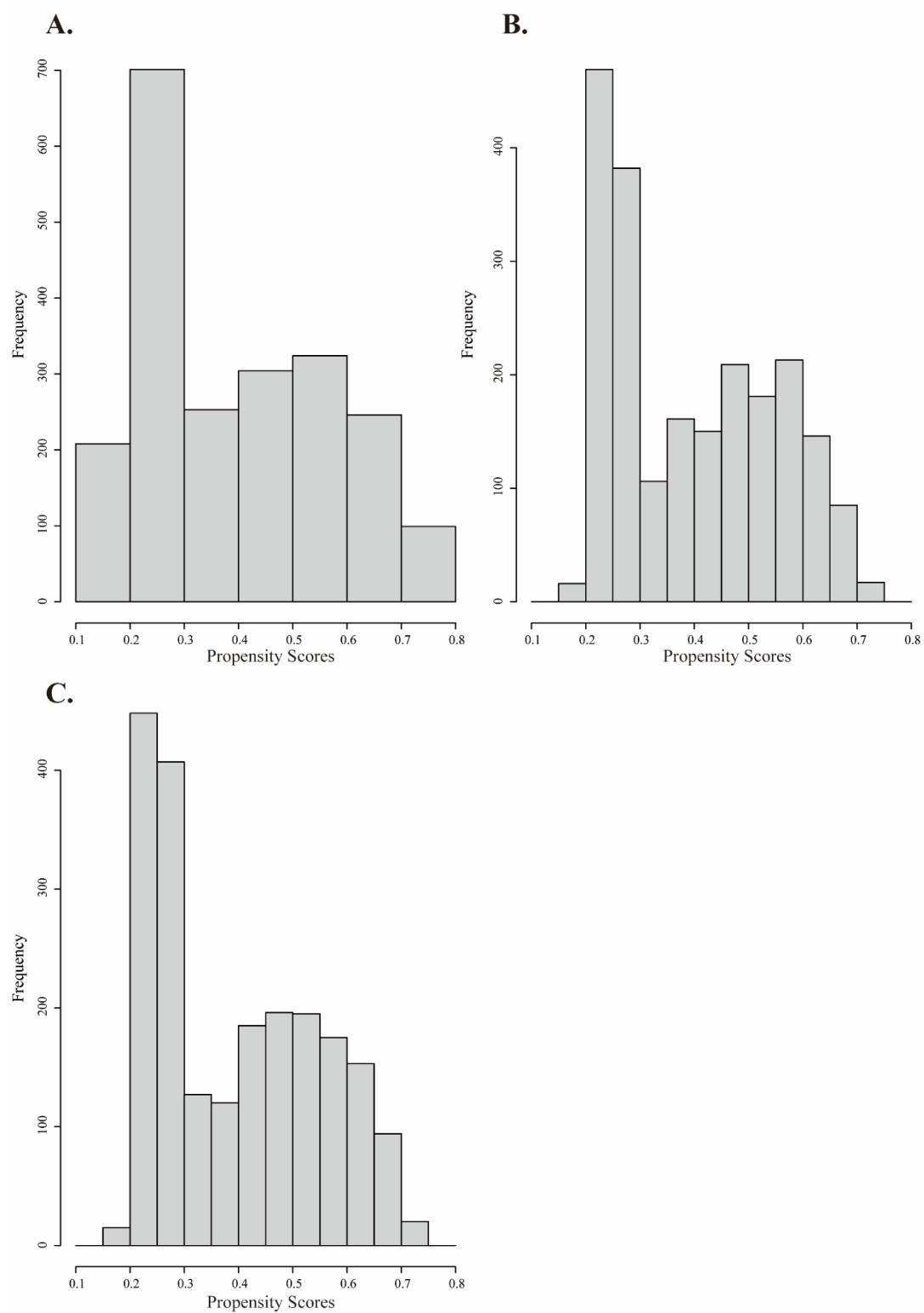
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Supplementary Figures

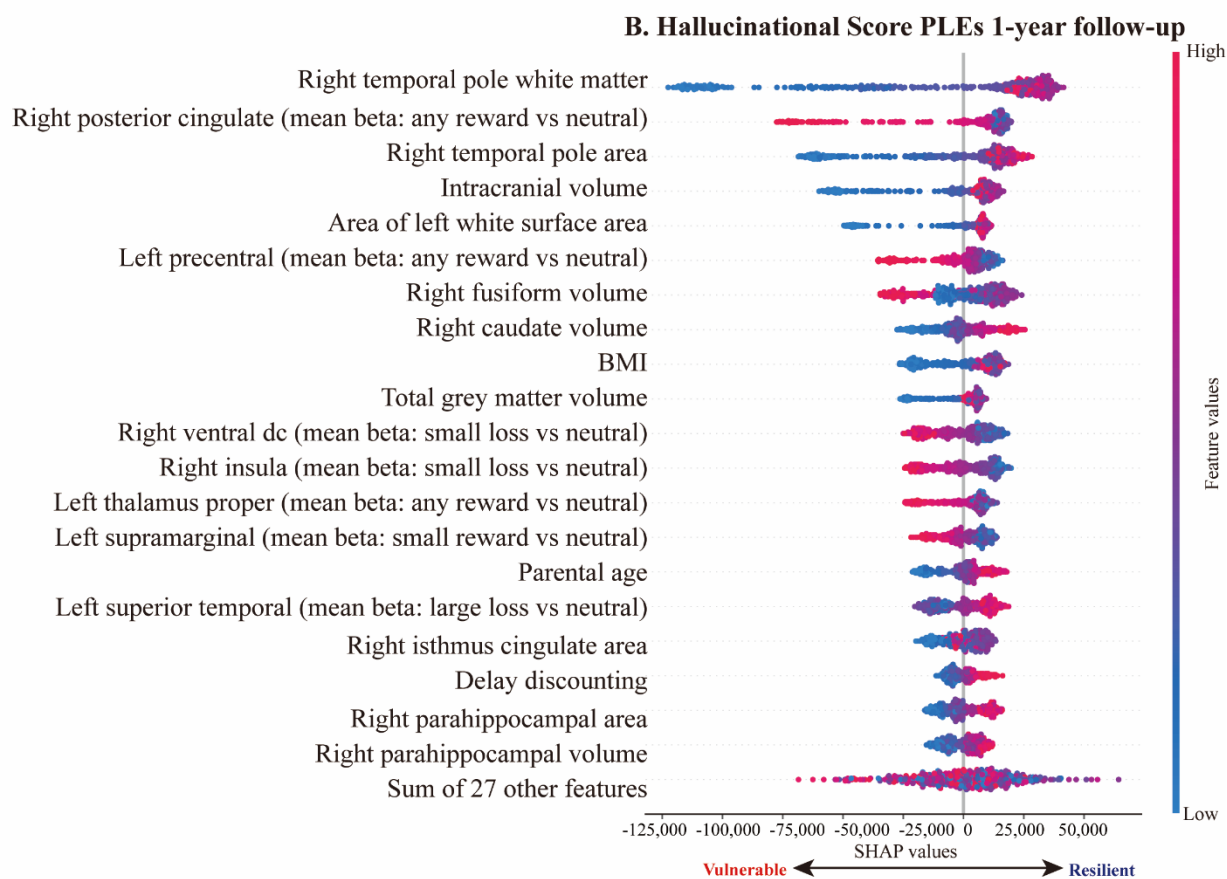
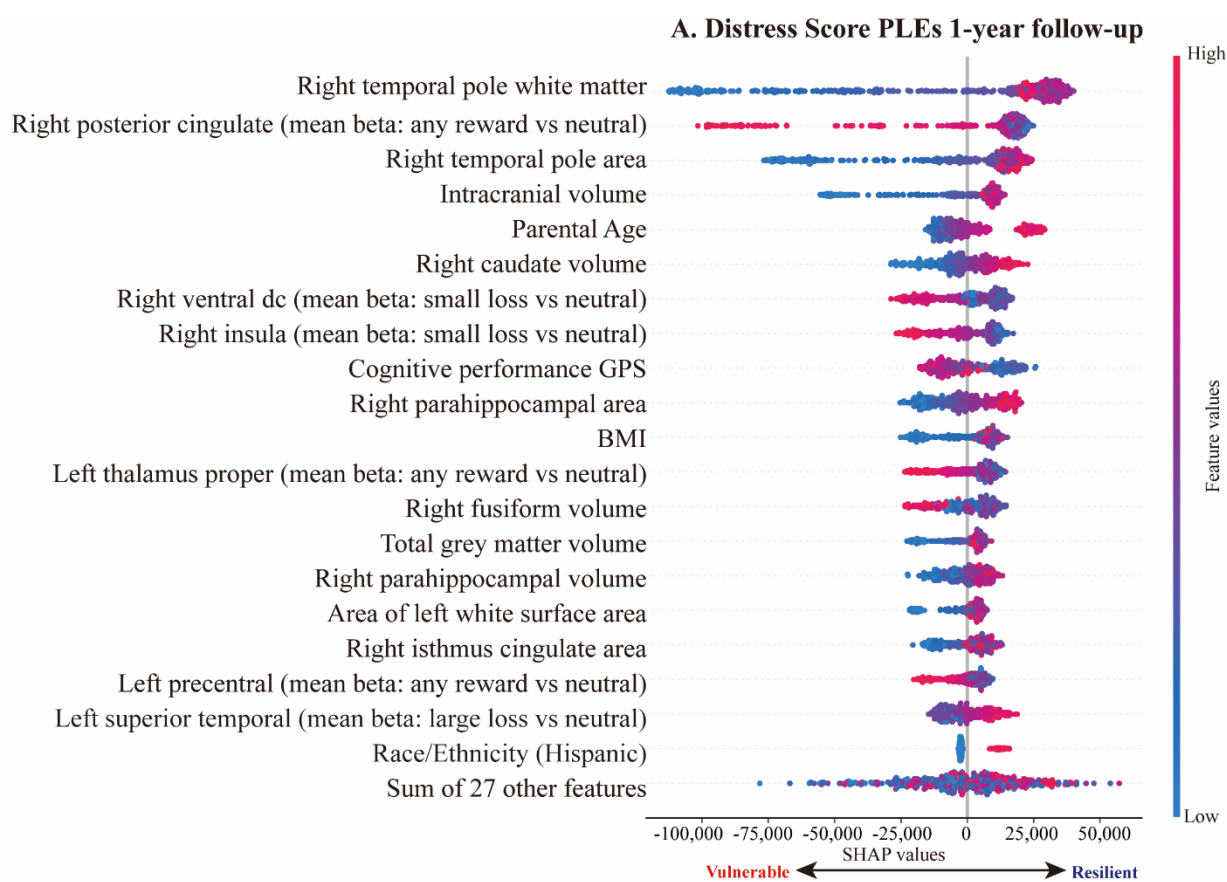
Fig. S1. Assessment of the overlap assumption. Histograms of propensity scores for the three distinct IV Forest models: Delay Discounting model (**A**), Gene-Brain model (**B**), and Integrated model (**C**). The IV Forest algorithm relies on the overlap assumption, which posits that there should be sufficient overlap in the covariate distributions between the treated and control groups. In other words, this means that the treatment status of an individual should not be deterministically predictable based on their covariates. The validity of the overlap assumption is evaluated using the histograms of propensity scores (65, 66). The fact that none of the estimated propensity scores are extremely close to either one or zero indicates that the overlap assumption holds, suggesting an appropriate level of randomness in treatment assignment relative to the covariates.



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Fig. S2. Beeswarm summary plots of Shapley additive explanation (SHAP) values for Integrated model. Contributions of the top 20 variables of highest importance in the Integrated model for the heterogeneous associations of neighborhood socioeconomic deprivation with 1-year follow-up distress score PLEs (**A**) and 1-year follow-up hallucinational PLEs (**B**) are shown. Variables are ordered by their relative importance in the model. Negative SHAP values indicate greater vulnerability (lower resilience) to the relationships of ADI with PLEs; Positive values indicate lower vulnerability (greater resilience). Contrasts of average beta activations of the given brain ROIs during MID tasks are shown in parenthesis. GPS: genome-wide polygenic scores; Ventral dc: ventral diencephalon.



Extended Discussion: Social Science & Economic Implications

Our finding that neighborhood socioeconomic adversity relates to steeper discounting of future rewards has profound implications for understanding the persistence of poverty and inequality. We propose three possible interpretations to explain the link between neighborhood socioeconomic adversity and individual's intertemporal decision-making, focusing on (i) individual's rate of delay discounting, (ii) resource scarcity, and (iii) social trust.

Challenging the Exogeneity of Time Preference

The longstanding economic theory posits that an individual's rate of discounting future rewards (time preference) is an exogenous parameter of intertemporal choice, established a priori, and impervious to external influences (24). Since the introduction of the discounted utility model (155) by Paul Samuelson in 1937, there has been limited exploration into whether environmental factors affect the development of an individual's parameter (24, 88). Our study challenges this notion by offering concrete evidence that the development of an individual's time preference is subject to environmental influences, and thereby opening new avenues for understanding the dynamics of intertemporal decision-making.

The Cognitive Impact of Resource Scarcity

Our second interpretation focuses on the cognitive consequences of poverty. Living in a state of resource scarcity can impose a significant “mental bandwidth tax,” overloading cognitive capacity and diverting attention away from long-term planning (156, 157). This cognitive strain can precipitate a range of poor financial decisions, such as impulsive purchasing or the mismanagement of finances, as immediate needs overwhelm future-oriented goals (157, 158). In this view, steeper discounting is a rational response to a cognitively demanding environment where long-term planning is a luxury.

The Role of Social Trust

The third interpretation emphasizes the role of social trust and environmental reliability. In neighborhoods characterized by instability and reduced social trust, promises of future rewards may be perceived as unreliable. When individuals lack confidence that a delayed reward will actually materialize, it is logical to prefer a smaller, certain reward available now over a larger, uncertain one in the future (159, 160). This learned preference for immediate gratification is an adaptive strategy in an untrustworthy environment.

Synthesis: The Behavioral Poverty Trap

These three interpretations, though seemingly distinct, converge in real-world contexts where socioeconomically disadvantaged families often face both resource scarcity and reduced social trust (161). This synthesis forms the basis of the “behavioral poverty trap”, wherein individuals raised in impoverished environments are prone to overvalue immediate rewards, leading to myopic behaviors such as overconsumption, inadequate savings (22, 157), and heightened risk of psychiatric disorders including psychosis and addiction (25, 162). These behaviors, in turn, may perpetuate socioeconomic challenges and hinder escape from poverty (20, 21).

Extended Discussion: Plausible Biological Mechanisms

We build on the behavioral poverty trap framework by identifying the potential link between neighborhood environment with intertemporal choice, leveraging longitudinal observations of preadolescent children aged 9-12 years, a critical period for neurocognitive development. A plausible biological mechanism for this phenomenon is the effects of glucocorticoid on brain's reward system. Prior studies indicate that adverse social environments induce chronic stress to children, elevating glucocorticoid hormones like cortisol (79-83). In particular, neighborhood socioeconomic deprivation has a more pronounced association with cortisol increases in children compared to any other social environmental factors (163). Long-term chronic stress from growing up in disadvantaged neighborhoods could result in epigenetic modifications affecting the mesocorticolimbic dopaminergic system, thereby altering the reward system (82, 83). This alteration may lead to a heightened preference for immediate rewards and impulsive behaviors, such as unhealthy eating and substance abuse (81-83, 164-168), further entrenching the cycle of socioeconomic disadvantage.

Our findings extend this understanding by linking the heterogeneous associations between ADI and children's PLEs with the intricate relationship between childhood social adversity and the reward system. Our findings suggest that these differential relationships of neighborhood socioeconomic adversity are modulated by genetic predispositions and neurodevelopmental traits associated with delay discounting. Children who experience residential deprivation and are at a higher PLEs demonstrate several distinct characteristics, including lower BMI, younger parental age, and altered brain structures and functions associated with delay discounting. Notably, these children showed reduced volume or white matter in specific brain regions (right temporal pole, right parahippocampal gyrus, right caudate nucleus, right isthmus cingulate),

1428 along with a smaller intracranial and total grey matter volume. Functionally, these children
1429 showed greater activation during MID tasks in regions including the right posterior cingulate,
1430 right ventral diencephalon, right insula, left precentral gyrus, left thalamus proper, and left
1431 superior temporal. This is particularly pronounced in children with a greater propensity for
1432 hallucinatory symptoms, who also show increased activity in the left supramarginal gyrus.

1433 It appears that variations in structural and functional aspects of the limbic system (the
1434 posterior cingulate, ventral diencephalon, insula, temporal pole, parahippocampal gyrus, and
1435 isthmus cingulate) play a crucial role in how socioeconomic hardship affects PLEs. This
1436 individual variability may be linked to individual differences in the glucocorticoid and reward
1437 system. The interaction between our genes and neurons, in response to chronic stress from poor
1438 socioeconomic conditions, may determine the differing associations of such adversity with PLEs.

1439 Although direct testing of this association within the ABCD Study samples was not feasible
1440 due to lack of relevant data, extensive animal and human corroborate our hypotheses. These
1441 studies suggest that maladaptive valuation of intertemporal rewards, namely the excessive
1442 discounting of future rewards, is linked to dysfunction of the prefrontal-limbic system, associated
1443 with psychopathologies such as psychosis in adolescents and adults (25, 26, 33, 84-87). Animal
1444 models have demonstrated that adverse social environments may trigger chronic dysregulation
1445 of glucocorticoid signaling in the hypothalamic-pituitary-adrenal axis and the dopaminergic
1446 mesocortical circuit (83), through epigenetic control (82, 83). This dysregulation could disrupt
1447 the adolescent reward circuit. In humans, childhood exposure to social adversity relates to
1448 changes in the hypothalamic-pituitary-adrenal axis and may contribute to psychosis through
1449 abnormal neurodevelopment of the limbic regions, the temporal pole, cingulate cortices,
1450 parahippocampal gyrus, and caudate nucleus (169-172). Young adults with a history of childhood
1451 social deprivation often show impaired reward processing, particularly in the cingulate and

mesostriatal dopaminergic system (19, 172-174).

The age of our study's participants, 9-12 years old, is a critical period for development of the prefrontal-limbic system (172, 175, 176). Children with psychotic disorders often exhibit greater reductions in grey matter compared to their healthy peers (177, 178). These neurodevelopmental alterations are associated with increased neuronal excitation, reduced inhibitory neural activities, and the resultant impulsive behaviors (179). In line with our findings, previous research has shown a correlation between higher PLEs and neuroanatomical alterations in the right temporal fusiform, right temporal pole, and right parahippocampal gyrus (171), as well as greater neural activations in limbic regions such as the insula and cingulate cortices during reward outcomes in MID task (180). Overall, our findings on the heterogeneous associations of neighborhood deprivation contribute to the growing body of literature showing the role of glucocorticoid and reward systems in modulating the negative relationships of environmental deprivation with psychosis (170, 172, 174, 181, 182).

Extended Discussion: Limitations and Future Directions

Several limitations of this study warrant consideration. First, we used ABCD Study, a non-randomized, observational cohort. Despite employing IV methods, including IV Forest and DoubleML, to adjust for both observed and potential unobserved confounders, the inherent limitation of the exclusion restriction assumption persists. This assumption, critical to the validity of the IV methods, cannot be directly verified with data. Albeit we substantiated this assumption with extensive prior research discussed in the **Methods** and the **Supplementary Methods**, its validity may still be subject to scrutiny, as might the overall efficacy of the IV method in fully adjusting for residual confounding bias.

For instance, SOI laws might influence child PLEs through pathways other than neighborhood adversity (ADI), such as by reducing parental stress associated with housing insecurity or by fostering a greater sense of community social capital. While our extensive covariates partially mitigate these concerns, we acknowledge that these unmeasured factors could potentially violate the exclusion restriction assumption. Future research could aim to measure these constructs directly to further disentangle these complex relationships.

To mitigate this limitation, we calculated E-values for the average effects of neighborhood disadvantage on delay discounting and PLEs. The large E-values calculated indicate that it would require unobserved confounders with a significantly strong association with both the exposure and the outcomes to negate our findings. Given the magnitude of these E-values and our comprehensive adjustments for confounding, it is unlikely that unobserved confounding could fully account for the observed relationships, thereby supporting the potential relationships found in this study, despite not providing absolute proof of causality.

Second, large number of samples were excluded from the initial cohort to ensure high data

quality for our neurobiological and behavioral analyses. As detailed in the **Methods** and **Supplementary Methods**, our commitment to data validity required the strict application of MRI quality control and behavioral validation criteria recommended by the ABCD consortium. While this approach maximizes the internal validity of our findings by ensuring we only modeled high-quality data, these necessary exclusions led to the substantial sample attrition and resulting selection bias discussed previously. This reflects a fundamental trade-off in large-scale cohort research between data rigor and sample representativeness. Consequently, while our findings are based on valid data, they must be interpreted with the understanding that they may not generalize to populations that were disproportionately excluded by these essential quality control procedures.

Third, since the majority of participants identified their race/ethnicity as white (65.57%), the generalizability of our findings to other minor race/ethnicity might remain to be tested. Nonetheless, recent research suggests that temporal discounting measures are consistent across diverse populations worldwide (61 countries, N=13,629) (183), which may mitigate concerns regarding the representativeness of our findings.

Fourth, the relatively short follow-up periods in our study (1-year and 2-year follow-up) may not adequately capture the long-term neurodevelopmental processes underlying intertemporal valuation and related psychopathology. Notably, additional follow-up data from the ABCD Study became available after we finalized this manuscript. As the ABCD Study continues to collect more longitudinal observations, longer follow-up periods in future studies could yield deeper insights.

Fifth, despite efforts to ensure representativeness by recruiting from diverse school systems across 21 research sites in the United States, our sample does not fully mirror the entire US

population (184). Though we provide a supplementary table (**Table S7**) comparing the demographic characteristics of our final sample with the general United States population, the generalizability of our specific neuro-genetic findings to other racial and ethnic groups remains to be tested.

Lastly, future research should examine the heterogeneous associations of additional environmental risk factors—such as parenting behavior (102) and early life stress (99)—as primary exposures to elucidate their potential effects on psychiatric disorders. Investigating how genetic and neural correlates interact with these risk factors will also advance our understanding of their unique contributions to individual differences in psychopathology.

1527 **Supplementary Tables**

1528 **Table S1. Exploratory analysis of psychopathologies associated with delay discounting.** As an exploratory analysis to find the which symptom is

1529 specifically associated with delay discounting among various psychopathologies, we conducted partial correlation analysis. PLEs were derived from the

1530 Prodromal Questionnaire-Brief Child Version (53) and other psychiatric symptoms were assessed with the Child Behavior Checklist (CBCL) (185). We used

1531 clinical diagnosis of each symptom based on the CBCL t-scores. These included internalizing problems, externalizing problems, total problems, and CBCL

1532 DSM-5 scales of depression, anxiety disorders, oppositional defiant, conduct problems, and attention deficit/hyperactivity disorder (ADHD). Estimates of

1533 Spearman ρ and multiple corrected p-values using false discovery rate are presented.

		Spearman ρ	t-statistic	p-FDR
PLEs	Distress Score PLE (1-year follow-up)	-0.058	-2.808	0.035
	Distress Score PLE (2-year follow-up)	-0.067	-3.246	0.024
	Delusional Score PLE (1-year follow-up)	-0.064	-3.091	0.024
	Delusional Score PLE (2-year follow-up)	-0.057	-2.761	0.035
	Hallucinational Score PLE (1-year follow-up)	-0.043	-2.057	0.191
	Hallucinational Score PLE (2-year follow-up)	-0.035	-1.684	0.318

CBCL	Internalizing Problems (1-year follow-up)	-0.015	-0.728	0.807
	Internalizing Problems (2-year follow-up)	-0.005	-0.221	0.971
	Externalizing Problems (1-year follow-up)	-0.003	-0.165	0.971
	Externalizing Problems (2-year follow-up)	0.013	0.634	0.838
	Total Problems (1-year follow-up)	-0.015	-0.721	0.807
	Total Problems (2-year follow-up)	0.012	0.584	0.838
	DSM-5 Depression (1-year follow-up)	-0.019	-0.920	0.793
	DSM-5 Depression (2-year follow-up)	-0.009	-0.445	0.927
	DSM-5 Anxiety Disorder (1-year follow-up)	-0.018	-0.848	0.793
	DSM-5 Anxiety Disorder (2-year follow-up)	-0.003	-0.139	0.971
	DSM-5 Somatic Problems (1-year follow-up)	0.003	0.143	0.971
	DSM-5 Somatic Problems (2-year follow-up)	-0.022	-1.058	0.774
	DSM-5 ADHD (1-year follow-up)	2.2.E-04	0.011	0.999
	DSM-5 ADHD (2-year follow-up)	0.018	0.886	0.793

	DSM-5 Oppositional Defiant (1-year follow-up)	-0.035	-1.682	0.318
	DSM-5 Oppositional Defiant (2-year follow-up)	3.68.E-05	0.002	0.999
	DSM-5 Conduct Problems (1-year follow-up)	-0.022	-1.061	0.774
	DSM-5 Conduct Problems (2-year follow-up)	0.007	0.346	0.971

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1541 **Table S2. E-values for Average Effects from IV Forest.** We calculated E-values for the average effects estimated from the IV Forest model. The E-values
1542 quantify the minimal strength of association that unobserved confounders, conditional on the observed covariates, would require with both the exposure (ADI)

1543 and the outcomes (delay discounting or PLEs) to potentially nullify the observed relationships (68).

	E-Values	E-value for Upper 95% CI
Delay Discounting (1-year follow-up)	9.13	1.87
Distress Score PLEs (1-year follow-up)	10.46	3.1
Distress Score PLEs (2-year follow-up)	7.32	2.08
Delusional Score PLEs (1-year follow-up)	457.03	1.89
Delusional Score PLEs (2-year follow-up)	77.22	1.52
Hallucinational Score PLEs (1-year follow-up)	60.79	1.24
Hallucinational Score PLEs (2-year follow-up)	155.27	5.12

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1547 **Table S3. Results of linear IV regression.** Conventional linear IV regression was conducted to confirm the average effects of neighborhood socioeconomic
1548 adversity on children’s delay discounting and PLEs. All p-values were corrected for multiple comparison using false discovery rate.

		Conventional Linear IV Regression				
		Estimates	Std. Error	95% Lower CI	95% Upper CI	P-FDR
Delay Discounting		-0.468	0.216	-0.891	-0.045	0.03
PLEs	Distress Score PLE (1-year follow-up)	0.609	0.224	0.170	1.047	0.011
	Distress Score PLE (2-year follow-up)	0.780	0.234	0.320	1.239	0.003
	Delusional Score PLEs (1-year follow-up)	0.486	0.215	0.066	0.906	0.028
	Delusional Score PLEs (2-year follow-up)	0.578	0.223	0.141	1.015	0.013
	Hallucinational Score PLEs (1-year follow-up)	0.604	0.221	0.172	1.036	0.011
	Hallucinational Score PLEs (2-year follow-up)	0.827	0.241	0.354	1.299	0.003

Double ML models

To confirm robustness of the IV Forest results, we used double machine learning (Double ML). This up-to-date causal machine learning method can

1555 utilize any state-of-the-art machine learning models to obtain consistent IV-based estimates of average effects by partialling out the confounding effects of
1556 covariates (70). It is particularly effective when the covariates are high-dimensional and have complex interactions.

1557 We used partial linear model instrumental variable model. In the partial linear model, we only assume linearity of the treatment variable ADI while the
1558 relationships between the outcome variable Y and covariates X and between instrumental variable SOI and covariates X remain as an unknown function.
1559 Below shows the simple mathematical representation of the model:

1560
$$Y = \mu ADI + g(X) + \varepsilon, \quad SOI = m(X) + v \quad (1)$$

1561 Here, Y denote for outcome variable (in our study, delay discounting and psychotic-like experiences), ADI the treatment variable, X multidimensional
1562 covariates, SOI the instrumental variable. $g(\bullet), m(\bullet)$ are unknown functions and ε, v are random errors. In the partial linear model (Equation 1), we assume
1563 that the treatment variable (i.e., ADI) have linear relationship with the outcome variable Y. There are no model assumptions specifying the relationship
1564 between multidimensional covariates X, outcome Y, and the instrumental variable SOI.

1565 We built an ensemble machine learning pipeline consisting of elastic net, random forest, XGBoost, and support vector machine with parameters tuned
1566 via 10-fold cross validation. In general, ensemble methods can improve model performance with lower error and higher accuracy by combining several base
1567 models (186). For each analysis, all continuous variables were standardized (z-scaled) beforehand to obtain standardized estimates, and analyses were run
1568 using *DoubleML* packages (69) in R version 4.1.2.

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1570 **Table S4. Results of Double ML models.** All p-values were corrected for multiple comparison using false discovery rate.

	Double ML Partial Linear IV Regression				
	Estimates	Std. Error	95% Lower CI	95% Upper CI	P-FDR
Delay Discounting (1-year follow-up)	-0.429	0.206	-0.834	-0.025	0.044
Distress Score PLE (1-year follow-up)	0.495	0.199	0.105	0.884	0.023
Distress Score PLE (2-year follow-up)	0.609	0.193	0.232	0.987	0.005
Delusional Score PLEs (1-year follow-up)	0.393	0.201	-0.001	0.788	0.051
Delusional Score PLEs (2-year follow-up)	0.417	0.196	0.032	0.801	0.044
Hallucinational Score PLEs (1-year follow-up)	0.498	0.187	0.131	0.864	0.018
Hallucinational Score PLEs (2-year follow-up)	0.683	0.188	0.316	1.051	0.002

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1572 **Random Forest-based Feature Selection using Boruta algorithm**

1573 We used Boruta (71) to select GPS and brain ROIs of structural MRI and MID task fMRI significantly associated with delay discounting. Boruta first
1574 generates shadow attributes, which is irrelevant to the outcome, by shuffling all input features. It then confirms features that have significantly higher

1575 importance in predicting the outcome than the shadow attributes with 95% confidence level, Bonferroni-corrected two-tailed tests (71). The selected features
1576 were included as covariates in the IV Forest models for assessing heterogeneous relationships of ADI.
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1578 **Table S5. Genetic and neural correlates of delay discounting selected using random forest-based algorithm.** Using the Boruta algorithm, we
1579 selected polygenic scores, neuroanatomical, and MID task functional activations related to delay discounting. meanImp, medianImp, minImp, and maxImp
1580 denote mean, median, maximal, and minimal importance, respectively. normHits indicate the number of hits normalized to number of importance source runs
1581 performed. *Decision* indicates the final decision of whether or not to select each feature.

		Boruta Feature Selection					
		meanImp	medianImp	minImp	maxImp	normHits	Decision
GPS	Cognitive performance GPS	9.219	9.344	1.540	14.803	0.989	Confirmed
	Educational attainment GPS	3.941	3.862	0.426	7.543	0.868	Confirmed
	IQ GPS (2-year follow-up)	6.798	6.806	1.975	9.841	0.995	Confirmed
Structural MRI	Left rostral anterior cingulate area	6.414	6.475	-0.035	9.673	0.986	Confirmed
	Area of the left white surface area	4.947	4.993	-0.651	8.320	0.966	Confirmed
	Right isthmus cingulate area	3.047	3.018	-1.526	6.367	0.748	Confirmed

	Right parahippocampal area	3.540	3.596	-0.736	7.584	0.848	Confirmed
	Right pars opercularis area	2.536	2.551	-2.733	5.521	0.597	Confirmed
	Right temporal pole area	3.646	3.642	-1.886	6.744	0.857	Confirmed
	Right fusiform volume	4.830	4.821	-0.220	8.109	0.960	Confirmed
	Right parahippocampal volume	2.619	2.596	-1.650	6.457	0.644	Confirmed
	Right pars opercularis volume	4.904	4.939	-2.184	8.229	0.969	Confirmed
	Right caudate volume	2.941	2.911	-1.467	6.777	0.703	Confirmed
	Total grey matter volume	5.226	5.237	-0.895	8.808	0.977	Confirmed
	Right Temporal pole white matter	6.748	6.804	0.408	9.862	0.987	Confirmed
MID task fMRI	Left precentral (mean beta: any reward vs neutral)	8.542	8.546	-0.835	12.054	0.997	Confirmed
	Left supramarginal (mean beta: any reward vs neutral)	8.795	8.818	-0.407	13.136	0.997	Confirmed
	Right posterior cingulate (mean beta: any reward vs neutral)	10.082	10.104	-0.002	13.946	0.997	Confirmed
	Left transverse temporal (mean beta: large reward vs neutral)	4.030	4.046	-1.268	7.737	0.901	Confirmed
	Left supramarginal (mean beta: small reward vs neutral)	7.297	7.319	0.857	11.480	0.997	Confirmed
	Left superior temporal (mean beta: large loss vs neutral)	2.781	2.782	-1.255	6.827	0.683	Confirmed
	Right insula (mean beta: small loss vs neutral)	3.419	3.437	-1.186	7.658	0.826	Confirmed
	Left thalamus proper (mean beta: any reward vs neutral)	7.081	7.078	-0.079	11.137	0.997	Confirmed
	Right ventral diencephalon (mean beta: small loss vs	4.407	4.397	-1.777	9.689	0.939	Confirmed

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1590 **Table S6. Evaluation of individual differences in the potential associations of neighborhood socioeconomic adversity with children’s PLEs.** We
1591 employed three distinct tests to evaluate the heterogeneous associations of ADI with PLEs: monotonicity test, alternative hypothesis test, and ANOVA test.
1592 These tests were applied to three developed IV Forest models: Delay Discounting model, Gene-Brain model, and Integrated model. All p-values were corrected
1593 for multiple comparison using false discovery rate. A star (*) denotes overall significance, indicating that a model passed all three heterogeneity tests.

		Monotonicity	Alternative Hypothesis				ANOVA		Overall Significance
		p-FDR	Estimate	Std. Error	t-value	p-FDR	F-value	p-FDR	
Delay Discounting Model	Distress Score PLEs (1-year follow-up)	0.037	-22.261	40.164	-0.554	0.799	1.114	0.48	
	Distress Score PLEs (2-year follow-up)	0.094	-54.769	142.528	-0.384	0.799	1.081	0.48	
	Delusional Score PLEs (1-year follow-up)	0.019	55.047	188.400	0.292	0.799	0.965	0.495	
	Delusional Score PLEs (2-year follow-up)	0.303	23.289	91.550	0.254	0.799	0.852	0.568	
	Hallucinational Score PLEs (1-year follow-up)	0.362	-86.756	241.597	-0.359	0.799	0.995	0.495	
	Hallucinational Score PLEs (2-year follow-up)	0.098	-119.595	341.060	-0.351	0.799	1.010	0.495	
Gene-Brain Model	Distress Score PLEs (1-year follow-up)	0.058	-9.522	4.244	-2.244	0.056	2.289	0.027	
	Distress Score PLEs (2-year follow-up)	0.099	-9.202	7.016	-1.312	0.285	1.608	0.161	
	Delusional Score PLEs (1-year follow-up)	0.086	-7.463	3.262	-2.288	0.056	2.631	0.01	
	Delusional Score PLEs (2-year follow-up)	0.043	-9.512	4.886	-1.947	0.094	3.493	0.001	
	Hallucinational Score PLEs (1-year follow-up)	0.022	-8.917	4.123	-2.163	0.062	1.878	0.083	
	Hallucinational Score PLEs (2-year follow-up)	0.146	-9.382	5.681	-1.651	0.162	2.695	0.009	
Integrated Model	Distress Score PLEs (1-year follow-up)	0.011	-8.922	2.922	-3.053	0.025	8.388	<0.001	*
	Distress Score PLEs (2-year follow-up)	0.074	-9.343	3.469	-2.693	0.025	7.361	<0.001	
	Delusional Score PLEs	0.172	-6.946	2.460	-2.824	0.025	6.816	<0.001	

	(1-year follow-up)								
	Delusional Score PLEs (2-year follow-up)	0.136	-6.854	2.617	-2.619	0.027	7.525	<0.001	
	Hallucinational Score PLEs (1-year follow-up)	0.038	-8.246	2.834	-2.910	0.025	7.182	<0.001	*
	Hallucinational Score PLEs (2-year follow-up)	0.089	-8.355	3.070	-2.721	0.025	9.484	<0.001	

Table S7. Comparison of the sample and national demographics. Since household income for subjects in the study is presented as deciles, it is transformed into a monetary value by considering the income limits for each decile. The data for the US national demographics is available at [Data is available at https://www.census.gov/en.html](https://www.census.gov/en.html).

	Mean or Ratio (%)
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		Subjects in our study	US National Demographics
Household Income		\$116,538	\$60,336
Sex	Male	53.86%	51.16%
Ethnicity/Race	White	65.57%	57.8%
	Black	6.37%	12.1%
	Hispanic	17.47%	18.7%
	Asian	0.33%	6.1%
	Other	10.26%	12.4%

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1604 **Adjusting for Sample Selection Bias with Inverse Probability Weighting**

1605 Our primary analysis required stringent quality control, resulting in a final analytical sample of 2,135 participants from an initial cohort of 11,876. To address
1606 potential selection bias stemming from this substantial sample reduction, we performed a sensitivity analysis using inverse probability weighting.

1607 This procedure involved two stages. First, we fit a logistic regression model to the full cohort (N = 11,876) to estimate each participant's probability of being

1608 included in the final sample. The predictors for this model comprised all baseline covariates used in our main Integrated model, including sociodemographics,
1609 delay discounting, genetic, and neuroimaging variables. Second, for each of the 2,135 participants in our final sample, we calculated stabilized weights as the
1610 inverse of their estimated probability of inclusion.

1611 Finally, we re-ran our primary IV Forest analysis, incorporating these stabilized weights into the estimation. This generated selection-bias-adjusted estimates
1612 of both the average and heterogeneous effects of neighborhood adversity.

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1614 **Table S8. Results of Sensitivity Analysis with Inverse Probability Weighting.** This table presents the results of a sensitivity analysis using inverse
1615 probability weighting to assess the impact of potential sample selection bias. The heterogeneous associations of neighborhood adversity (ADI) with 1-year
1616 follow-up distress and hallucinational PLEs, estimated from our Integrated model, remained statistically significant after the inverse probability weighting
1617 adjustment.

		Monotonicity	Alternative Hypothesis				ANOVA		Overall Significance
		p-FDR	Estimate	Std. Error	t-value	p-FDR	F-value	p-FDR	
Integrated Model	Distress Score PLEs (1-year follow-up)	0.011	-8.922	2.922	-3.053	0.008	8.388	<0.001	*
	Distress Score PLEs	0.074	-9.343	3.469	-2.693	0.008	7.361	<0.001	

	(2-year follow-up)								
	Delusional Score PLEs (1-year follow-up)	0.172	-6.946	2.460	-2.824	0.008	6.816	<0.001	
	Delusional Score PLEs (2-year follow-up)	0.136	-6.854	2.617	-2.619	0.009	7.525	<0.001	
	Hallucinational Score PLEs (1-year follow-up)	0.038	-8.246	2.834	-2.910	0.008	7.182	<0.001	*
	Hallucinational Score PLEs (2-year follow-up)	0.089	-8.355	3.070	-2.721	0.008	9.484	<0.001	

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